

The middle and the ends II.

Vertex crossings in a symmetric Markov multinomial model

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Abstract

A ball falls through a Galton board with d directions. It picks its first direction uniformly, and at each later peg it keeps its direction with probability p and otherwise turns to one of the other $d - 1$ directions at random. For $d = 2$ this is the persistent Galton board of Kagey's Problem 131, and for $d = 3$ it is his tetrahedral board. The bin is the vector of counts of the d directions, a point of a simplex. When p is close to 1 the most likely bins are the vertices, which only a ball that never turns can reach. We ask when the most likely bin on each face of the simplex becomes as likely as a vertex. Put $T = N(1 - p)/(d - 1)$ for N steps. To first order every face catches up at the same place, $T \approx \log N$. We break this tie at second order. The central bin of a face with k directions crosses a vertex when $T + \frac{1}{2} \log T = \log N + a_k + 1/(8T) + O((\log N)^{-2})$, with explicit constants $a_2 > a_3 > \dots$, each the mass of a face over a Gaussian volume. So for large N the mode jumps once, from the vertices straight to the center. A start law or a weak field shifts the constants and can let an intermediate face win.

Keywords: persistent random walk; Galton board; Potts chain; Markov multinomial distribution; occupation counts; mode; spanning trees; Bessel functions.

MSC2020: 60J10; 60F05; 60F10; 82B20; 05C05; 33C10.

1 Introduction

1.1 The question and the answer

Problem 131 of Kagey's Open Problem Collection [1] is about a Galton board on which the ball tends to keep going the same way. At the first peg the ball goes left or right with probability $\frac{1}{2}$, and at every later peg it repeats its previous direction with probability p . Kagey asks when the middle bin is exactly as likely as the 2 end bins, and Paper A [2] answers this. Kagey also asks what changes on a tetrahedron. There each peg sends the ball in one of 3 directions instead of 2, and the bins after N steps form a triangle, one layer of the tetrahedron.

In this paper the ball has any number $d \geq 2$ of directions. It picks its first direction uniformly. At each later peg it keeps its direction with probability p , and otherwise it turns to one of the other $d - 1$ directions, each with probability $r = (1 - p)/(d - 1)$. After N steps its bin is the vector $k = (k_1, \dots, k_d)$ of the numbers of steps in each direction. So the bins are the points of a simplex: a segment for $d = 2$, Kagey's triangle for $d = 3$. The corners Ne_i of the simplex are its vertices. Only a ball that never turns reaches a vertex, so each vertex has probability p^{N-1}/d .

When p is close to 1 the ball rarely turns, and the vertices are the most likely bins. As p decreases the other bins catch up. On the triangle there are 2 natural candidates to catch up first:

the middle of an edge, where the ball has used 2 directions, and the middle of the triangle, where it has used all 3. Which one wins, and when?

Here is the answer. Put $L = \log N$ and $T = Nr$. Every face of the simplex catches up with the vertices at the same place to first order, when T is about L . So to first order the question has no answer. We break this tie at second order, and the center of the whole simplex wins. As p decreases, the most likely bin jumps from the vertices straight to the center, and for large N no edge or other proper face ever holds it. On the triangle the middle bin $(\frac{N}{3}, \frac{N}{3}, \frac{N}{3})$ overtakes the corners while the middles of the edges are still about 17% below them. Figure 2 shows this at $N = 24$.

Why every face ties. Take a face with k directions. A ball that uses only these directions must avoid the other $d - k$ directions at every peg, and it does so with probability $(1 - (d - k)r)^{N-1} \approx e^{-(d-k)T}$. So the face carries mass about $\frac{k}{d}e^{-(d-k)T}$, while a vertex carries $\frac{1}{d}e^{-(d-1)T}$. On the face the ball turns about T times towards each direction, so the fractions k_i/N fluctuate by about $1/\sqrt{T}$ around $1/k$. The mass of the face is therefore spread over about $(N/\sqrt{T})^{k-1}$ bins near its center, and the central bin gets a share of about $(\sqrt{T}/N)^{k-1}$. Altogether

$$\frac{\text{central bin of the face}}{\text{vertex}} \approx \text{const} \cdot \left(\frac{e^T \sqrt{T}}{N} \right)^{k-1}.$$

The bracket is the same for every face. It crosses 1 when $T + \frac{1}{2} \log T = L + O(1)$, at the same place for every face, and even the term $\frac{1}{2} \log T$ is shared. Only the constant in front can tell the faces apart.

How second order breaks the tie. The constant is $e^{-(k-1)a_k}$, with a_k given in (1) below. It has 2 sources. The face has k times the start mass of a vertex, and the fractions have Gaussian fluctuations with covariance Σ_k/T for an explicit matrix Σ_k . So $e^{-(k-1)a_k} = k \det(2\pi\Sigma_k)^{-1/2}$, the mass of the face divided by a Gaussian volume (Proposition 3.7). Now write the ratio as $\exp((k-1)[\Psi - a_k])$ with the common bracket $\Psi = T - L + \frac{1}{2} \log T$. As p decreases, T and Ψ increase. The constants decrease, $a_2 > a_3 > \dots > a_d$, so the full face, which has the smallest constant, reaches the vertices first, when $\Psi = a_d$. At that moment a face with $k < d$ directions is still below the vertices by the factor $e^{-(k-1)(a_k - a_d)}$. On the triangle this factor is $e^{-(a_2 - a_3)} = 0.827$, which is the 17% above. Before that moment every proper face is below the vertices, since $\Psi < a_d < a_k$. After it the full center is above every proper face, since $(d-1)(\Psi - a_d) > (k-1)(\Psi - a_k)$ when $\Psi \geq a_d$. So no proper face is ever the mode.

1.2 Setting and notation

We collect the notation here. Fix $d \geq 2$ and a start law μ on the set $[d] = \{1, \dots, d\}$ of directions, which we also call states. Let $X_1, X_2, \dots$ be the Markov chain with $X_1 \sim \mu$ that keeps its state with probability $p \in [0, 1]$ and moves to each of the other $d-1$ states with probability $r = (1-p)/(d-1)$. Put $\sigma = p - r = (dp - 1)/(d-1)$. Kagey's board is $d = 2$ with the uniform start.

Bins. The count vector $K_N = (K_{N,1}, \dots, K_{N,d})$ records how often each state occurs among $X_1, \dots, X_N$. A bin is a vector k of nonnegative integers with sum N , and $P_N^\mu(k) = \mathbb{P}(K_N = k)$. For the uniform start $\mu_i = 1/d$ we write $P_N(k)$. The uniform start is the default everywhere except in Section 4.

Faces and vertices. A face is a nonempty set $F \subseteq [d]$, and $\mu(F) = \sum_{a \in F} \mu_a$. A bin k lies on the face $\text{supp } k = \{i : k_i > 0\}$. The vertices are the faces of size 1, that is, the bins Ne_i . A

word that reaches Ne_i never switches, so $P_N^\mu(Ne_i) = \mu_i p^{N-1}$, and for the uniform start we write $V_N = p^{N-1}/d$.

Balanced bins. A bin with k positive coordinates is balanced if these coordinates differ by at most 1. For the uniform start all balanced bins with k positive coordinates have the same probability, since they differ only by a permutation of the states. We call it $C_{N,k}(p)$, so $C_{N,1} = V_N$.

Scales. Put $L = \log N$ and $T = Nr$. The expected number of switches is $(N-1)(1-p) = (d-1)T(1-1/N)$, so T is about the expected number of switches divided by the number $d-1$ of possible targets. Put $\tau = T/L$. For $p > 0$ let $u = r/p$, the odds of a switch to one given state against no switch, and $z = Nu = T/p$. For $p > 1/d$ we have $\sigma > 0$ and also use $v = r/\sigma = u/(1-u)$. Since $z - T = (d-1)Tz/N$, the 2 scales z and T differ by $O(L^2/N)$ when $T = O(L)$. This is smaller than every error term in our asymptotic statements, so these statements hold with either one. At finite N the 2 differ, and the heuristic Remark 4.7 says which one it uses. Exact statements are written in u .

Crossings. For $2 \leq k \leq d$, the crossing $p_{N,k}$ is the value of p at which $C_{N,k} = V_N$, and $z_{N,k} = N(1 - p_{N,k})/(d-1)$ is the value of T there (not of z). We write $p_{N,k}^{(d)}$ when the number of states matters.

The constants. For $k \geq 2$ put

$$a_k = \frac{1}{2} \log(4\pi) - \frac{k + \frac{1}{2}}{k-1} \log k, \quad D_k = e^{-(k-1)a_k} = \frac{k^{k+1/2}}{(4\pi)^{(k-1)/2}}, \quad \Psi_N(T) = T - L + \frac{1}{2} \log T. \quad (1)$$

Numerically $a_2 = -0.4674$, $a_3 = -0.6571$ and $a_4 = -0.8139$.

Matrix form. The transition matrix is $I + (T/N)Q$ with $Q = J - dI$, where J is the all-ones matrix, so Q switches at rate 1 to each other state. The exit rate of a face F with k states is $\lambda_F = d - k$, the principal Dirichlet eigenvalue of $-Q$ on F . Paper C [3] calls K_F the second-order constant of a face. Here $K_F = \mu(F)k^{k-1/2}(4\pi)^{-(k-1)/2}$ for $k \geq 2$ and $K_{\{i\}} = \mu_i$ (Section 4), and for the uniform start $a_k = -\frac{1}{k-1} \log(K_F/K_{\{i\}})$.

Potts chain. With the uniform start and $0 < p < 1$, a word $x_1 \cdots x_N$ has probability $\frac{1}{d} r^{N-1} (p/r)^{\#\{s < N: x_s = x_{s+1}\}}$. This is the one-dimensional d -state Potts model with free boundaries. For $d = 2$ it is the Ising chain of Paper A.

Asymptotic notation. In each statement the constants in $O(\cdot)$, and the threshold behind “for large N ”, depend only on the quantities declared fixed in that statement (always including d and k). Uniformity in the other variables is stated each time.

1.3 Main results

Theorem 3.1 makes the heuristic of Section 1.1 precise. Uniformly for $cL \leq T \leq CL$, the logarithm of $C_{N,k}/V_N$ is $(k-1)[\Psi_N(T) - a_k]$, up to a next term $-(k-1)/(8T)$ that is again proportional to $k-1$ and an error $O(T^{-2} + T^2/N)$. In the first-order language of Paper C, where $T = \tau L$ and only multiples of L count, a balanced bin on a face F has probability $\exp(-L[\tau\lambda_F + |F| - 1] + O(\log L))$. On the complete graph every face attains the minimum $d-1$ of $\tau\lambda_F + |F| - 1$ at $\tau = 1$, so the first-order theory cannot choose.

By Theorem 3.2, for every $N \geq k$ the balanced bin with k positive coordinates crosses the vertices at exactly one $p_{N,k}$, which increases with N , and

$$z_{N,k} + \frac{1}{2} \log z_{N,k} = L + a_k + \frac{1}{8z_{N,k}} + O(L^{-2}). \quad (2)$$

For $k = 2$ this is the crossing equation of Paper A, and $a_2 = \frac{1}{2} \log(\pi/8)$ is its constant c_0 . Since $a_2 > a_3 > \dots > a_d$, Corollary 3.9 shows that for large N and all $T \leq CL$ the global mode changes exactly once, from the d vertices straight to the balanced bins with d positive coordinates. All the face-center crossings lie in a p -interval of length $(d-1)(a_2 - a_d + o(1))/N$. For small N this can fail: on the triangle with $N = 4$ an edge center can be the mode (Example 3.10).

The same method gives the crossing of every bin whose positive coordinates are of order N (Theorem 3.3). For fractions α its constant is $D(\alpha) = A^2 / \sqrt{\det(2\pi\Sigma_\alpha)}$ with $A = \sum_i \sqrt{\alpha_i}$, where Σ_α is the occupation covariance of a tilted chain, and $\det \Sigma_\alpha$ is a weighted count of the spanning trees of the complete graph K_k (Proposition 3.6). At the center the count is Cayley's k^{k-2} , and $D(\alpha) = D_k$.

Then we perturb the model in 2 ways, each of which acts at order 1, the order of the constants a_k . A start law μ changes the constant of a face F against a vertex i to $a_k + \frac{1}{k-1} \log(k\mu_i/\mu(F))$ (Theorem 4.3). On the triangle with $\mu = (\frac{1}{2}, \frac{1}{2}, 0)$ this lets an edge hold the mode on a window of width $\log(16/(9\sqrt{3})) = 0.026$ in T (Remark 4.6). A Potts field of size h_i/N per visit to state i adds to each face the field averaged over the face (Theorem 5.2), and one field makes every support size win in turn (Corollary 5.6).

Finally we drop the assumption that the positive coordinates are of order N . A Bessel majorant has relative error $O(L^2/N)$ for every positive count vector when $T = O(L)$ (Theorem 6.2). When at least 2 coordinates are of order N , a small coordinate starts to matter once it has size N/L^2 (Theorem C.1). Near a vertex, with one large count b and small counts of total $o(b)$, the crossing odds are $u = \sqrt{y/b}(1 + o(1))$, where y is the root of a product of Laguerre-type polynomials (Theorem 6.4).

The sections follow this order: the exact law (Section 2), the tie (Section 3), the start, the field and the boundary (Sections 4 to 6), and the checks and open questions (Section 7). The longer proofs are in Appendices A to D.

1.4 The series

This is the second of 4 papers on Problem 131, with the common title “The middle and the ends” (I is Paper A, II is this Paper B, III is Paper C and IV is Paper D). They share the following notation.

- Kagey’s board has N steps (his row n has $N = n - 1$). The first step is left or right with probability $\frac{1}{2}$, and each later step repeats the previous one with probability p . Put $q = 1 - p$ and $u = q/p$, the odds of a switch.
- $P_N(k)$ is the probability of k right steps, that is, of bin k . The endpoints have $P_N(0) = P_N(N) = \frac{1}{2}p^{N-1}$.
- $L = \log N$ and $T = Nq$, which is the expected number $(N-1)q$ of switches up to an error below 1.
- The crossing is the value of p , or of T , at which a bin is exactly as likely as an endpoint. For the middle bin it is p_N , and $z_N = N(1 - p_N)$.
- In Papers B and C the chain has d states. A face F is a nonempty set of states, the vertices are the faces of size 1, λ_F is the principal Dirichlet eigenvalue of $-Q$ on F , and K_F is the second-order constant.

With d directions we replace q by $r = q/(d-1)$, so $T = Nr$ and $u = r/p$. For $d = 2$ these are the quantities above, $P_N(k, N-k)$ is $P_N(k)$, and $z_{N,2} = z_N$. Paper A [2] brackets z_N for every $N \geq 2$ with uniform Bessel bounds and proves its crossing equation with an explicit error. Paper C [3] gives the general theory for $P = I + (T/N)Q$, with the first-order selection of Section 1.3 and the

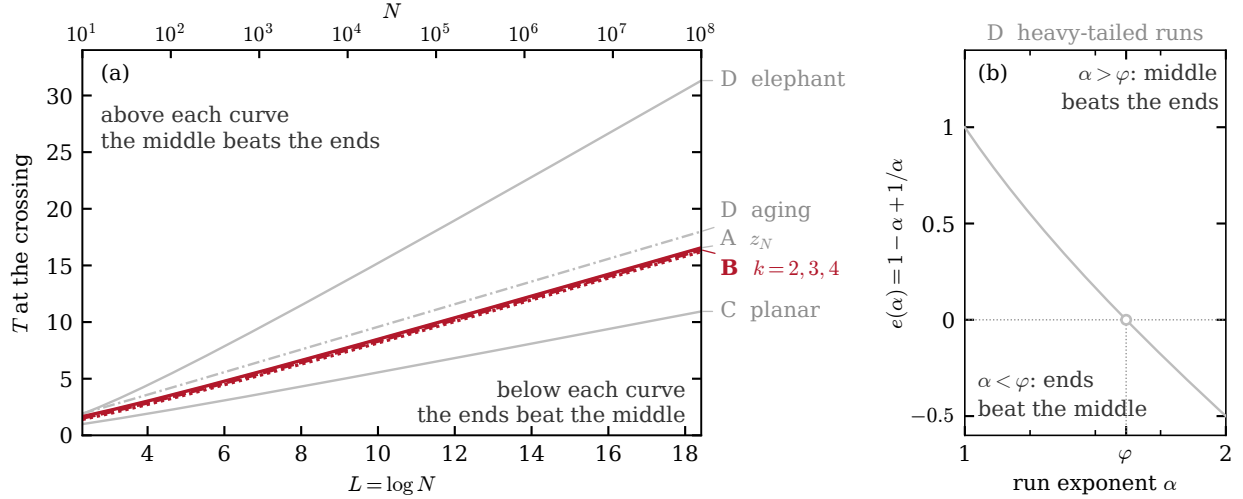


Figure 1: (a) The crossing on each model's switching scale against $L = \log N$ for Papers A to D. The scale is T for Papers A to C, $x = n\varepsilon$ for the elephant walk and the expected number of switches for the aging walk of Paper D. The bold curves are the face-center crossings of this paper for faces with $k = 2, 3, 4$ states. They nearly coincide at this scale, since $a_2 - a_4 < 0.35$. (b) The exponent $e(\alpha) = 1 - \alpha + 1/\alpha$ of Paper D for runs with heavy tails and a stationary start, which changes sign at the golden ratio φ .

second-order constants K_F . The present paper is the case where the first order is silent on every face, and it resolves the tie in full. Paper D [4] asks what memory changes. Figure 1 shows the crossings of all 4 papers.

1.5 Related work

The law belongs to the theory of Markov multinomial distributions. Wang and Yang [5] extended the Markov Bernoulli model of Edwards [6] to finitely many states. We have seen neither paper, and we describe both through Wang and Tang [7, 8]. The chain of Wang and Yang starts from a law μ and moves from i to j with probability $\theta\delta_{ij} + (1 - \theta)\mu_j$, and the generating function of the sum of its states is a product of matrices. Our chain with the uniform start is the case where μ is uniform and $\theta = \sigma$. So the exact law for the uniform start is a special case of this theory, and we do not claim it as new. Our general start changes only the initial law and keeps the uniform switching, so it is a Wang–Yang chain only when μ is uniform. Hsiao [9] and Wang and Tang [8] prove compound Poisson limits for sums over multistate chains. Whittle [10] gave the exact law of the transition counts of a finite chain. The count vector is determined by the transition counts and the first state, so P_N could also be summed from his formula.

Brydges, van der Hofstad and König [11] give occupation densities for continuous-time chains on the visited faces of the simplex. Huang, Kious, Sidoravicius and Tarrés [12, Theorem 1] give the joint density of the local times, the last-exit spanning tree and the net crossing numbers of a walk with symmetric conductances, with Bessel functions. So in continuous time the trees appear directly, but we do not use their formula. Stadje [13] gives generating functions for the joint numbers of visits of a discrete chain and limit laws when one state is nearly absorbing. Dietz and Sethuraman [14] study chains that switch from i to j at time n with probability $G(i, j)/n$, where every $G(i, j) > 0$. Their chains make about $\log N$ switches by time N but only $O(1)$ of them

after time $N/2$, and their occupation fractions have a limit law with full support on the simplex, which is a Dirichlet law when $G(i, j)$ depends only on j . Near our crossings the chain makes about $(d - 1) \log N$ switches, spread evenly in time. Dupuis and Liu [15] give the rate function of the occupation measure, which we compare with our exponent in Section 3. Kassan-Ogly and Filippov [16] solve the Potts chain in a field in the thermodynamic limit. For Whittle, Hsiau, Stadje and Kassan-Ogly and Filippov we have read only the abstracts, and we use nothing beyond them. None of these works compare one count vector with a vertex when p depends on N , and we did not find the face-center crossings or the tie in the literature. We take the Bessel expansions from [18].

2 The model and the exact law

This section writes P_N^μ in 3 forms, each from a different way of reading a word $X_1 \cdots X_N$. The *run form* counts the switches. It gives all our exact statements, such as the uniqueness of each crossing. The *refresh form* reads the chain in another way. Its transition matrix is $\sigma I + rJ$, so when $p \geq 1/d$ the chain keeps its state with probability σ and otherwise draws a fresh state uniformly from all d states, possibly the same one. The *ratio form* is the refresh form divided by a vertex. For $p > 1/d$ it is a sum of positive terms, and all our asymptotics start from it. The notation is that of Section 1.2.

For a positive count vector $n = (n_1, \dots, n_k)$ with total N and a state a , let $W_n^{(a)}(j)$ be the number of words with counts n and j changes that start with a , and let $W_n(j) = \sum_a W_n^{(a)}(j)$. Put

$$S_n^{(a)}(u) = \sum_j W_n^{(a)}(j) u^j, \quad S_n(u) = \sum_a S_n^{(a)}(u) = \sum_j W_n(j) u^j.$$

These polynomials depend only on the positive coordinates, and not on d . For a bin k with d coordinates we use them with n the vector of positive coordinates of k .

Proposition 2.1 (the generating function). *Put $y_i = x_i t / (1 - \sigma x_i t)$ and $P_0^\mu(0, \dots, 0) = 1$. As formal power series,*

$$\sum_{N \geq 0} \sum_{k_1 + \dots + k_d = N} P_N^\mu(k) x_1^{k_1} \cdots x_d^{k_d} t^N = 1 + \frac{\sum_i \mu_i y_i}{1 - r \sum_i y_i}.$$

Proof. Let F_i be the generating function of the words of length at least 1 that end in state i , each weighted by its probability, and let $S = \sum_i F_i$. Conditioning on the last letter gives $F_i = x_i t (\mu_i + p F_i + r(S - F_i)) = x_i t (\mu_i + \sigma F_i + rS)$, that is, $F_i = y_i (\mu_i + rS)$. Summing over i gives $S = \sum_i \mu_i y_i + rS \sum_i y_i$, and the generating function is $1 + S$. $\square$

Proposition 2.2 (the exact law). *Let $N \geq 1$, let k have nonnegative integer coordinates with sum N , and put $F = \text{supp } k$. Sums over m run over the integer vectors with $m_i = 0$ for $i \notin F$ and $1 \leq m_i \leq k_i$ for $i \in F$, and $M = \sum_i m_i$.*

(a) (*refresh form*) With $0^0 = 1$,

$$P_N^\mu(k) = \sum_m \left(\sum_{i \in F} \mu_i m_i \right) \frac{(M-1)!}{\prod_i m_i!} r^{M-1} \sigma^{N-M} \prod_{i \in F} \binom{k_i - 1}{m_i - 1}.$$

For the uniform start this reads

$$P_N(k) = \frac{1}{d} \sum_m \frac{M!}{\prod_i m_i!} r^{M-1} \sigma^{N-M} \prod_{i \in F} \binom{k_i - 1}{m_i - 1}.$$

(b) (run form) Let $p > 0$ and let n be the vector of positive coordinates of k . Then $P_N^\mu(k) = p^{N-1} \sum_{a \in F} \mu_a S_n^{(a)}(u)$. In particular $P_N(k)/V_N = S_n(u)$.

(c) (ratio form) If $1/d < p < 1$, so that $0 < u < 1$ and $v > 0$, then

$$S_n(u) = (1-u)^{N-1} \sum_m M! v^{M-1} \prod_{i \in F} \frac{1}{m_i!} \binom{n_i-1}{m_i-1}, \quad (3)$$

which is a sum of positive terms.

When $\sigma \geq 0$, the refresh form has a direct reading. Cut the word into M blocks at the $M-1$ refreshes. Each refresh lands on a prescribed state with probability r , and each of the other $N-M$ steps keeps the state with probability σ . If m_i blocks carry state i , there are $m_i(M-1)!/\prod_j m_j!$ orders of the block labels that start with i , and $\binom{k_i-1}{m_i-1}$ ways to choose the lengths of the m_i blocks of state i . The proof below does not need $\sigma \geq 0$.

Proof. (a) By the proof of Proposition 2.1, $S = \sum_{M \geq 1} r^{M-1} (\sum_i \mu_i y_i) (\sum_i y_i)^{M-1}$. By the multinomial theorem the coefficient of $\prod_i y_i^{m_i}$ in $(\sum_i \mu_i y_i) (\sum_i y_i)^{M-1}$ is $\sum_i \mu_i m_i (M-1)!/\prod_j m_j!$. With $w_i = x_i t$ we have $y_i = w_i/(1 - \sigma w_i)$, and $[w^k](w/(1 - \sigma w))^m = \binom{k-1}{m-1} \sigma^{k-m}$ for $k \geq m \geq 1$. For $m = 0$ only $k = 0$ contributes. Multiplying the coefficients proves (a), and for the uniform start $\sum_i \mu_i m_i = M/d$.

(b) A word that starts with a and has j changes has probability $\mu_a p^{N-1-j} r^j = \mu_a p^{N-1} u^j$. Sum over the words with counts k .

(c) Divide the uniform case of (a) by $V_N = p^{N-1}/d$. Since $r/p = u$ and $\sigma/p = 1 - u$, we have $r^{M-1} \sigma^{N-M}/p^{N-1} = u^{M-1} (1-u)^{N-M} = (1-u)^{N-1} v^{M-1}$. $\square$

At $p = 1/d$ we have $\sigma = 0$ and only $M = N$ survives, which gives the multinomial law for the uniform start. At $p = 1$ only $M = 1$ survives, which gives mass μ_i at $N e_i$. For $\sigma < 0$ the terms in (a) can cancel, so for numerical work we then use exact arithmetic or dynamic programming.

2.1 Bessel bounds

For $x \geq 0$ the modified Bessel function of order 1 is $I_1(x) = \sum_{j \geq 0} (x/2)^{2j+1}/(j!(j+1)!)$. Since $\binom{2j+1}{j} \leq 4^j$, comparing coefficients gives $I_1(x) \leq \sinh x \leq e^x$. We also use the expansion [18, Eq. 10.40.1]

$$I_1(x) = \frac{e^x}{\sqrt{2\pi x}} \left(1 - \frac{3}{8x} + O(x^{-2})\right), \quad x \rightarrow \infty,$$

and the function

$$\Phi(x) = \sum_{j \geq 0} \frac{x^j}{j!(j+1)!} = \frac{I_1(2\sqrt{x})}{\sqrt{x}}, \quad \Phi(0) = 1.$$

Since $\binom{2j}{j} \leq 4^j$, we have $1/(j!(j+1)!) \leq 4^j/(2j)!$, and comparing term by term with the even part of the series of $e^{2\sqrt{x}}$ gives $\Phi(x) \leq e^{2\sqrt{x}}$. Lemmas A.1, A.2 and A.3 in Appendix A give the other facts about Φ that we use, a bound for products, and an expansion of Gaussian averages. The comparison of Laguerre-type polynomials with Φ that we need near the boundary is Lemma 6.1.

3 The tie and its resolution

In this section the start is uniform. We prove the tie of Section 1.1 and break it (Subsection 3.1), write the constants as Gaussian volumes and spanning-tree sums (Subsection 3.2), and find the global mode (Subsection 3.3). The constants a_k , D_k and Ψ_N are those of (1). Let n be a balanced vector with k positive coordinates and total N . By Proposition 2.2(b), $C_{N,k}/V_N = S_n(u)$, and as a function of u this ratio does not depend on d .

3.1 Every face ties at first order

Theorem 3.1 (every face ties at first order). *Fix $d \geq 2$ and $0 < c < C$.*

(i) *For large N , uniformly for $cL \leq T \leq CL$ and $2 \leq k \leq d$,*

$$\log \frac{C_{N,k}}{V_N} = (k-1)[\Psi_N(T) - a_k] - \frac{k-1}{8T} + O\left(\frac{1}{T^2} + \frac{T^2}{N}\right).$$

(ii) *Put $T = \tau L$. Uniformly for $c \leq \tau \leq C$ and $1 \leq k \leq d$, a face F with k states has*

$$\log C_{N,k} = -L[\tau \lambda_F + k - 1] + O(\log L), \quad \lambda_F = d - k.$$

At $\tau = 1$ the right side is $-(d-1)L + O(\log L)$ for every k .

(iii) *Put $T = L - \frac{1}{2} \log L + t$. Uniformly for t in a compact interval, for each $2 \leq k \leq d$,*

$$\frac{C_{N,k}}{V_N} \longrightarrow e^{(k-1)(t-a_k)}. \quad (4)$$

(iv) $a_2 > a_3 > \dots$.

In words, the central bin of every face reaches the vertices when $\Psi_N(T)$ is of order 1, that is, when $T = L - \frac{1}{2} \log L + O(1)$. Part (ii) is the first-order statement: on the scale L all faces have the same exponent at $\tau = 1$. Part (iii) zooms into the window where the tie lives. There each ratio has a limit, and the faces differ only through the constants a_k . Paper C describes such tie windows in general, and Remark 5.7 explains why among symmetric kernels only the uniform one ties on every face.

Theorem 3.2 (face-center crossings). *For every $2 \leq k \leq d$ and $N \geq k$ there is exactly one $p = p_{N,k} \in (0, 1)$ with $C_{N,k}(p) = V_N(p)$. The balanced bin is more likely than a vertex below $p_{N,k}$ and less likely above it. The crossing $p_{N,k}$ is strictly increasing in N ,*

$$p_{k,k} = \frac{1}{1 + (d-1)(k!)^{-1/(k-1)}},$$

and every crossing has $u \leq (k!)^{-1/(k-1)} < 1$, so $p_{N,k} > 1/d$. For fixed k the value $z_{N,k} = N(1 - p_{N,k})/(d-1)$ of T at the crossing satisfies

$$z_{N,k} + \frac{1}{2} \log z_{N,k} = L + a_k + \frac{1}{8z_{N,k}} + O(L^{-2}),$$

and

$$z_{N,k} = L - \frac{1}{2} \log L + a_k + \frac{\frac{1}{4} \log L - \frac{1}{2} a_k + \frac{1}{8}}{L} + O\left(\frac{(\log L)^2}{L^2}\right). \quad (5)$$

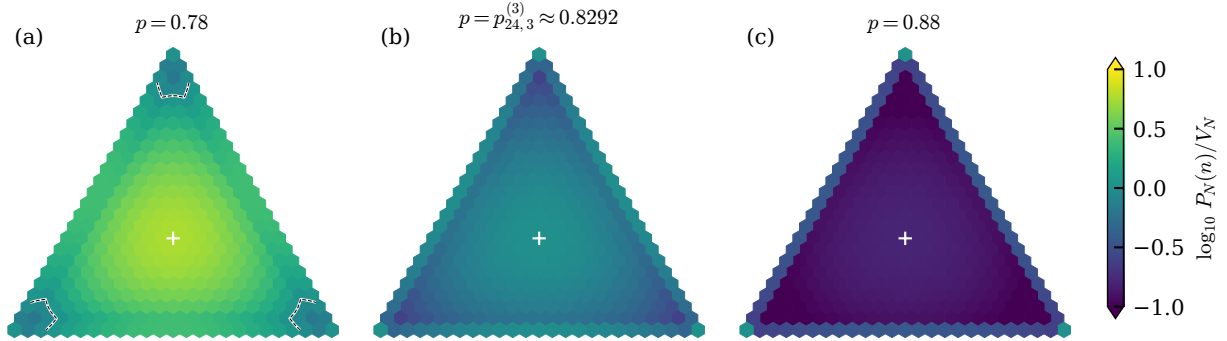


Figure 2: The ratio $P_N(n)/V_N$ for $d = 3$ and $N = 24$, one hexagon per bin, on a $\log_{10}$ scale. (a) At $p = 0.78$ only the 21 bins inside the dashed curves near the vertices are below the vertex height. (b) At the full-center crossing $p_{24,3}^{(3)} = 0.82921 \dots$ the center $(8, 8, 8)$ (cross) ties with the vertices and every other bin is below them. (c) At $p = 0.88$ every nonvertex bin is below the vertex height.

In words, each face center crosses the vertices exactly once, the crossing moves towards $p = 1$ as N grows, and the crossing equation is the same for every face except for the constant a_k . For $d = 2$, $z_{N,2} = N(1 - p_N)$ is the crossing z_N of [2], and $a_2 = \frac{1}{2} \log(\pi/8)$ is its constant c_0 . The proofs here do not use the binary case. Figure 2 shows the full-center crossing on the triangle at $N = 24$.

Idea of the proofs. The proofs are in Appendix A. The exact part uses the run form. The polynomial S_n has nonnegative coefficients, and its first nonzero coefficient is $W_n(k-1) = k!$, from the words that use each state in a single run. So S_n increases from 0 to ∞ and crosses 1 exactly once. Its next coefficient $W_n(k) = k!(k-1)(N-k)/2$ grows with N , which moves the root. The asymptotic part uses the ratio form (3). This sum is close to a Bessel integral (Theorem 6.2), and the Laplace expansion of this integral (Lemma A.4) gives, for a balanced n with k positive coordinates and uniformly for $cL \leq z \leq CL$,

$$S_n(z/N) = D_k \frac{z^{(k-1)/2} e^{(k-1)z}}{N^{k-1}} \left(1 - \frac{k-1}{8z} + O\left(\frac{1}{z^2} + \frac{z^2}{N}\right) \right). \quad (6)$$

This is the heuristic of Section 1.1 with all constants. Then $\log V_N = -(d-1)T - \log d + O(T^2/N)$ and $z - T = O(L^2/N)$ give parts (i) to (iii) and the expansions of the crossing. Part (iv) follows from the monotonicity of $(x + \frac{1}{2}) \log x / (x - 1)$.

3.2 The constants as Gaussian volumes

The heuristic of Section 1.1 says that D_k should be the mass of a face divided by the Gaussian volume of the fluctuations of the fractions. We now make this precise in 2 ways. First we find the crossing at every interior point of a face and write its constant with a covariance matrix and a spanning-tree sum (Proposition 3.6). Then we sum over the whole face (Proposition 3.7). We evaluate the constants at the actual fractions $\alpha_i = n_i/N$, since we do not assume that the count vector converges at any rate.

Theorem 3.3 (interior compositions). *Fix $k \geq 2$, $\varepsilon > 0$ and $0 < c < C$. Let $n_1 + \dots + n_k = N$ with $\alpha_i = n_i/N \geq \varepsilon$, and put*

$$A = \sum_{i=1}^k \sqrt{\alpha_i}, \quad g = A^2 - 1, \quad s = k - 1, \quad D(\alpha) = \frac{A^{k/2+1}(\prod_i \alpha_i)^{-3/4}}{(4\pi)^{s/2}}.$$

Uniformly for these count vectors and $cL \leq z \leq CL$,

$$S_n(z/N) = D(\alpha) \frac{z^{s/2} e^{gz}}{N^s} \left(1 + O_{k,\varepsilon,C}(z^{-1} + z^2/N) \right). \quad (7)$$

The unique positive root u_n of $S_n(u) = 1$ satisfies

$$Nu_n = \frac{s}{g}L - \frac{s}{2g} \log L - \frac{\log D(\alpha) + (s/2) \log(s/g)}{g} + O_{k,\varepsilon} \left(\frac{\log L}{L} + \frac{L^2}{N} \right). \quad (8)$$

For a fixed target α and counts $n_i = N\alpha_i + O(1)$, the constants may be evaluated at the target without changing the error order.

In words, a bin with fractions α grows like $e^{g(\alpha)z}$ against a vertex, so it crosses at $z \approx (s/g)L$. By Cauchy–Schwarz $g \leq k - 1$, with equality only at the center, so on each face the center crosses first, at $z \approx L$. For $k = 2$, $g = 2\sqrt{\alpha_1\alpha_2}$. At the center $A = \sqrt{k}$, $g = k - 1$ and $D(\alpha) = D_k$, which gives the leading term of (6). Since $z - T = O(L^2/N)$, the right side of (7) may also be written with T in place of z , and (8) also holds with the value of T at the root in place of Nu_n .

The exponent g is a drop in a rate function. By Dupuis and Liu [15, Eq. (3.2)], the occupation measure of the chain with generator $Q = J - dI$ has rate function $I(\alpha) = -\langle \sqrt{\alpha}, Q\sqrt{\alpha} \rangle = d - (\sum_i \sqrt{\alpha_i})^2$. So $g(\alpha) = I(e_1) - I(\alpha)$, the drop in the rate from a vertex to α . Paper C proves that the inverse Hessian of the rate is the occupation covariance of a tilted chain, on every face where the chain is irreducible. So $D(\alpha)$ should be a Gaussian factor. We now show this directly and write it as a spanning-tree sum. The next lemma computes the determinant of an occupation covariance for any reversible chain.

Lemma 3.4 (occupation covariance). *Let G be an irreducible generator on $\{1, \dots, k\}$, reversible with respect to $\pi > 0$, with conductances $c_{ij} = \pi_i G_{ij} = c_{ji}$ for $i \neq j$. For $\theta \in \mathbb{R}^k$ let $f = \theta - \pi(\theta)\mathbf{1}$, and let χ solve $-G\chi = f$ with $\pi(\chi) = 0$. Let Σ be the matrix of the quadratic form $\theta \mapsto 2 \sum_i \pi_i f_i \chi_i$ on the vectors with $\theta_1 = 0$, in the coordinates $\theta_2, \dots, \theta_k$. Let $\mathcal{L}_{(1)}$ be the Laplacian $\mathcal{L} = -\text{diag}(\pi)G$ without its first row and column. Then*

$$\det \Sigma = \frac{2^{k-1}(\prod_i \pi_i)^2}{\det \mathcal{L}_{(1)}} = \frac{2^{k-1}(\prod_i \pi_i)^2}{\text{tree}(c)}, \quad \text{tree}(c) = \sum_{\mathcal{T}} \prod_{\{i,j\} \in \mathcal{T}} c_{ij},$$

where the sum is over the spanning trees $\mathcal{T}$ of the complete graph on $\{1, \dots, k\}$.

The form $2 \sum_i \pi_i f_i \chi_i$ is the usual asymptotic variance per unit time of $\int_0^t \theta(Y_s) ds$ for the chain Y with generator G , but we use it only as a definition. The proof in Appendix A writes Σ with $\mathcal{L}_{(1)}^{-1}$ and uses the matrix determinant lemma and the matrix-tree theorem [17]. For product conductances the tree sum has a closed form.

Lemma 3.5 (weighted Cayley formula). *For $x_1, \dots, x_k > 0$ and conductances $c_{ij} = x_i x_j$,*

$$\text{tree}(c) = x_1 x_2 \cdots x_k (x_1 + \cdots + x_k)^{k-2}.$$

With every $x_i = 1$ this is Cayley's count k^{k-2} of the spanning trees of K_k .

Proof. Put $X = \sum_i x_i$. The Laplacian is $\text{diag}(Xx_i) - xx^\top$, so $\mathcal{L}_{(1)} = \text{diag}(Xx_2, \dots, Xx_k) - x'x'^\top$ with $x' = (x_2, \dots, x_k)$. The matrix determinant lemma gives $\det \mathcal{L}_{(1)} = X^{k-1} \prod_{i \geq 2} x_i (1 - \sum_{i \geq 2} x_i/X) = X^{k-2} \prod_i x_i$, and this is $\text{tree}(c)$ by the matrix-tree theorem. For a second proof, note that a tree $\mathcal{T}$ has weight $\prod_i x_i^{\deg_{\mathcal{T}}(i)}$. The Prüfer code is a bijection from the labeled trees on $\{1, \dots, k\}$ to the words of length $k-2$ in which i occurs $\deg_{\mathcal{T}}(i) - 1$ times, so the sum is $\prod_i x_i \cdot X^{k-2}$. $\square$

Proposition 3.6 (the tree form of the constants). *Fix $k \geq 2$ and α in the open simplex. Put $h = \sqrt{\alpha}$ entrywise and $A = \sum_i h_i$.*

(1) *The chain G^α with rates $G_{ij}^\alpha = h_j/h_i$ for $i \neq j$ is reversible with stationary law α and conductances $h_i h_j$. Let $Q_k = J - kI$ and $\theta_\alpha = -(Q_k h)/h$ entrywise. Then $Q_k + \text{diag}(\theta_\alpha)$ is symmetric with Perron root 0 and Perron vector h , and G^α is its Doob transform $\text{diag}(h)^{-1}(Q_k + \text{diag}(\theta_\alpha))\text{diag}(h)$. It is not a Doob transform of Q_k itself unless α is uniform.*

(2) $\text{tree}(\alpha) := \sum_{\mathcal{T}} \prod_{\{i,j\} \in \mathcal{T}} \sqrt{\alpha_i \alpha_j} = (\prod_i \sqrt{\alpha_i}) A^{k-2}$.

(3) *With Σ_α the occupation covariance of G^α from Lemma 3.4, the constant of Theorem 3.3 is*

$$D(\alpha) = \frac{A^2}{(4\pi)^{(k-1)/2}} \frac{\sqrt{\text{tree}(\alpha)}}{\prod_i \alpha_i} = \frac{A^2}{\sqrt{\det(2\pi \Sigma_\alpha)}}.$$

(4) *At the center $G^\alpha = Q_k$, each of the k^{k-2} spanning trees has weight k^{1-k} , and $\text{tree}(\alpha) = 1/k$. In the coordinates $\alpha_2, \dots, \alpha_k$, $\Sigma_k = (2/k^2)(I - J/k)$ and $\det \Sigma_k = 2^{k-1} k^{1-2k}$. So*

$$D_k = \frac{k \cdot k^k}{(4\pi)^{(k-1)/2}} \sqrt{k^{k-2} \cdot k^{1-k}} = e^{-(k-1)a_k}.$$

(5) *The same holds for the constant $D_{k_1}(\beta)$ of Theorem C.1, with k_1 and β in place of k and α .*

The proof is a direct check, given in Appendix A. Here is how to read the factors in (3) for the uniform start. First, $A^2 = d(\mu \cdot h)(h \cdot 1)$, one factor for the start of the path and one for its free end, and Remark 4.2 explains the start factor. Next, $(4\pi)^{-(k-1)/2}$ is the Gaussian normalization $(2\pi)^{-(k-1)/2}$ times $2^{-(k-1)/2}$, from the 2 in the variance $2\langle f, (-G)^{-1} f \rangle_\pi$. And $\sqrt{\text{tree}(\alpha)}/\prod_i \alpha_i = 2^{(k-1)/2}(\det \Sigma_\alpha)^{-1/2}$ by Lemma 3.4. On other reversible faces this square root of a weighted tree sum is the form that survives [3].

The second way sums the whole face. Its total mass is explicit, and comparing it with the Gaussian sum of the interior profile gives D_k again.

Proposition 3.7 (mass balance). *Fix $k \geq 2$, and let the sums run over the positive vectors $n \in \mathbb{Z}^k$ with $n_1 + \dots + n_k = N$.*

(1) *For every $N \geq k$ and every u ,*

$$\sum_n S_n(u) = \sum_{j=1}^k (-1)^{k-j} \binom{k}{j} j (1 + (j-1)u)^{N-1}.$$

(2) *For fixed $0 < c < C$, uniformly for $cL \leq z \leq CL$ and $u = z/N$, this sum is $k e^{(k-1)z} (1 + o(1))$, and it is also $D_k (2\pi)^{(k-1)/2} (\det \Sigma_k)^{1/2} e^{(k-1)z} (1 + o(1))$.*

(3) Hence $D_k = k \det(2\pi\Sigma_k)^{-1/2}$.

Proof of (1). A word of length N over a set of q letters with exactly j changes is fixed by its first letter, the positions of the changes and the new letter at each change. So there are $q(q-1)^j \binom{N-1}{j}$ of them. By inclusion and exclusion over the set $G \subseteq \{1, \dots, k\}$ of allowed letters, the number of such words that use all k letters is $\sum_G (-1)^{k-|G|} |G|(|G|-1)^j \binom{N-1}{j}$, and this is $\sum_n W_n(j)$. Multiply by w^j and sum over j . $\square$

Parts (2) and (3) are proved in Appendix A. The term $j = k$ in (1) is $P_N(\text{supp } K_N \subseteq F)/V_N$ for a face F with k states, and it gives the leading term. In words, D_k is the mass k of the face, in units of a vertex, divided by the Gaussian volume $\det(2\pi\Sigma_k)^{1/2}$ of the occupation fluctuations. Proposition 3.7 is a consistency identity and an interpretation, not an independent derivation of a_k , since part (2) uses the shape of Theorem 3.3 near the center.

3.3 Balanced bins and the global mode

So far we have compared balanced bins with vertices. To find the global mode we also need that, within a face, the balanced bins are the best.

Theorem 3.8 (balancing within a support). *For $1/d \leq p < 1$, among bins with a prescribed set of positive coordinates and total N , the probability is largest exactly at the balanced bins. More precisely, moving one unit from a coordinate b to another coordinate a , where $b \geq a + 2$ and $a \geq 1$, strictly increases the probability of the bin.*

The reason is simple. Once the number of runs of each state is fixed, splitting the visits evenly gives the most ways to choose the run lengths. In the proof in Appendix A this becomes the log-concavity in n of the polynomials B_n of (17). The condition $p \geq 1/d$ holds at every crossing, by Theorem 3.2.

Corollary 3.9 (exactly one mode change). *Fix $d \geq 2$ and $C > 1$. There is N_0 such that for $N \geq N_0$ and every p with $0 < T \leq CL$ the following hold.*

- (i) *If $p > p_{N,d}$, the global modes are exactly the d vertices.*
- (ii) *If $p < p_{N,d}$, they are exactly the balanced bins with d positive coordinates.*
- (iii) *If $p = p_{N,d}$, they are these 2 classes and nothing else.*

Moreover $p_{N,2} < p_{N,3} < \dots < p_{N,d}$ and $p_{N,d} - p_{N,k} = (d-1)(a_k - a_d + o(1))/N$. So all face-center crossings lie in a p -interval of length $(d-1)(a_2 - a_d + o(1))/N$, and as p decreases the full center crosses first.

In words, for large N the mode jumps from the vertices straight to the center of the simplex. Here is the idea. By Theorem 3.8 every global mode is a vertex or a balanced bin. By Theorem 3.1(i) a balanced bin with k positive coordinates has log ratio $(k-1)(\Psi_N(T) - a_k) + o(1)$ to a vertex, and a_d is the smallest constant. So the full center reaches the vertices first, while every proper face is still below them, as explained in Section 1.1. The proof is in Appendix A. Corollary 3.9 extends the window statement (4) to all of $T \leq CL$. It needs N to be large.

Example 3.10 (an intermediate face can be a finite-size mode). Corollary 3.9 does not hold for every N . Take $d = 3$, $N = 4$ and $p = 10/17$, so $u = 7/20$. For an edge center and a full center,

$$\frac{P_4(2, 2, 0)}{V_4} = 2u + 2u^2 + 2u^3 = \frac{4123}{4000} > 1, \quad \frac{P_4(2, 1, 1)}{V_4} = 6u^2 + 6u^3 = \frac{3969}{4000} < 1.$$

Here $p > 1/3$, so by Theorem 3.8 the 3 edge centers are the only global modes. Also $0 < T = 14/17 < \log 4$, so this p lies in the range of Corollary 3.9 for every $C > 1$. Hence $N_0 > 4$ when $d = 3$.

4 A general start

Kagey's ball picks its first direction uniformly. Suppose instead that it starts in state a with probability μ_a , and moves as before. The start only affects the first run, so one might expect it to matter little. It does matter: it changes the constants a_k by an amount of order 1, which is exactly the order at which the faces are separated. So a start can decide which face wins.

For a count vector n with support F , Proposition 2.2(b) gives

$$P_N^\mu(n) = p^{N-1} \sum_{a \in F} \mu_a S_n^{(a)}(u) = p^{N-1} S_n(u) \sum_{a \in F} \mu_a \omega_a(n), \quad \omega_a(n) = \frac{S_n^{(a)}(u)}{S_n(u)}.$$

So the start enters only through the weights $\omega_a(n)$, which sum to 1. A vertex has $V_i^\mu = \mu_i p^{N-1}$. Throughout this section, F is a face with $k = |F| \geq 2$ states and $\mu(F) > 0$ (if $\mu(F) = 0$, every bin with support F has probability 0), and i is a vertex with $\mu_i > 0$.

Lemma 4.1 (start weights). *Let n be a positive count vector on F .*

- (1) *If $n_a = n_b$, then $\omega_a(n) = \omega_b(n)$. So $\omega_a(n) = 1/k$ for all $a \in F$ when n is balanced and k divides N .*
- (2) *Fix $C > 0$. There are C_1 and N_0 , depending only on d , k and C , such that for $N \geq N_0$, $0 < T \leq CL$ and every balanced bin n of F ,*

$$\left| \sum_{a \in F} \mu_a \omega_a(n) - \frac{\mu(F)}{k} \right| \leq \mu(F) C_1 \frac{(1+T)^2}{N}.$$

In words, a balanced bin forgets where the path started, up to a relative error $O(T^2/N)$. The proof is in Appendix B. For bins that are not balanced the weights are not uniform, and the next remark says what they are.

Remark 4.2 (where a conditioned path starts). Under the uniform start, $\omega_a(n)$ is the probability that the chain starts in a given $K_N = n$. Fix a compact set $\mathcal{K}$ in the open face and $0 < c < C$, and put $\alpha = n/N$ and $A = \sum_{b \in F} \sqrt{\alpha_b}$. Then $\omega_a(n) = \sqrt{\alpha_a}/A + O(1/L)$, uniformly for $\alpha \in \mathcal{K}$ and $cL \leq T \leq CL$ (proof omitted; checked numerically, see Section 7). So a path conditioned to have fractions α starts in a with probability close to $\sqrt{\alpha_a}/A$, and not α_a . In the notation of Proposition 3.6, $\sqrt{\alpha}$ is the Perron vector h of the tilted generator.

Theorem 4.3 (face-center crossings with a start). *Let n be a balanced bin of F .*

- (1) For every $N \geq k$, the ratio $P_N^\mu(n)/V_i^\mu$ is a polynomial in u with nonnegative coefficients, zero constant term and lowest term $(\mu(F)/\mu_i)(k-1)!u^{k-1}$. So it equals 1 at exactly one $p \in (0,1)$, and the bin is more likely than the vertex below this p . If k divides N , then $P_N^\mu(n) = \frac{\mu(F)}{k} p^{N-1} S_n(u) = \frac{d\mu(F)}{k} P_N(n)$. So the crossing solves $S_n(u) = k\mu_i/\mu(F)$, its u decreases strictly along $N \in k\mathbb{Z}$, and at $N = k$ it is $u = (\mu_i/(\mu(F)(k-1)!))^{1/(k-1)}$.
- (2) Fix $C > 0$ and let N_0 be as in Lemma 4.1(2). For $N \geq N_0$, uniformly for $0 < T \leq CL$ and over all μ , $P_N^\mu(n) = \frac{\mu(F)}{k} p^{N-1} S_n(u)(1 + O((1+T)^2/N))$.
- (3) Put $a_k(\mu) = a_k + \frac{1}{k-1} \log \frac{k\mu_i}{\mu(F)}$. Fix $0 < c < C$. For large N , uniformly for $cL \leq T \leq CL$ and over all μ ,

$$\log \frac{P_N^\mu(n)}{V_i^\mu} = (k-1)[\Psi_N(T) - a_k(\mu)] - \frac{k-1}{8T} + O\left(\frac{1}{T^2} + \frac{T^2}{N}\right).$$

So the value $z = z_N^\mu$ of T at the crossing satisfies $z + \frac{1}{2} \log z = L + a_k(\mu) + \frac{1}{8z} + O(L^{-2})$ and the expansion (5) with $a_k(\mu)$ in place of a_k . With $T = L - \frac{1}{2} \log L + t$ the ratio tends to $e^{(k-1)(t-a_k(\mu))}$, uniformly on compact sets of t .

In words, a start changes only the constant of each crossing. It lowers a_k by $\frac{1}{k-1} \log(\mu(F)/(k\mu_i))$, the logarithm of the average start weight on the face against the start weight of the vertex, shared among the $k-1$ new states. The error terms do not depend on μ .

Proof. (1) Each $S_n^{(a)}$ has nonnegative coefficients, zero constant term and lowest term $(k-1)!u^{k-1}$, from the $(k-1)!$ orders of the states that start with a , with each state in a single run. So the ratio increases strictly from 0 to ∞ in u , and $u = r/p$ decreases from ∞ to 0 on $(0,1)$. If k divides N , Lemma 4.1(1) gives $\sum_a \mu_a \omega_a(n) = \mu(F)/k$. The rest of (1) is about S_n , which increases strictly along $N \in k\mathbb{Z}$ at every $u > 0$ and equals $k!u^{k-1}$ at $N = k$, by the proof of the exact part of Theorem 3.2.

(2) is Lemma 4.1(2), divided by $\mu(F)/k$.

(3) By (2), $\log(P_N^\mu(n)/V_i^\mu) = \log S_n(u) - \log(k\mu_i/\mu(F)) + O(T^2/N)$, and $S_n(u) = C_{N,k}/V_N$. So Theorem 3.1(i) gives the display. The rest follows as in the proof of Theorem 3.2, since only the constant has changed. $\square$

In the notation of Paper C [3], where K_F is the second-order constant of a face, $K_F = \mu(F)k^{k-1/2}(4\pi)^{-(k-1)/2}$ and $K_{\{i\}} = \mu_i$ here, so $a_k(\mu) = -\frac{1}{k-1} \log(K_F/K_{\{i\}})$. So K_F depends on the start only through $\mu(F)$, and the uniform start gives $a_k(\mu) = a_k$.

If μ is constant on F , say $\mu_a = m$ for $a \in F$, then $P_N^\mu(n) = m p^{N-1} S_n(u)$ for every n with support in F . So the statements of Sections 3 and 6 about such bins hold verbatim, with V_N replaced by $V_a^\mu = m p^{N-1}$ for a vertex $a \in F$. In particular Theorem 3.8 holds on F . Otherwise balancing can fail.

Example 4.4 (balancing can fail). Let $d = 2$, $N = 4$ and $\mu = (\frac{1}{2} + \varepsilon, \frac{1}{2} - \varepsilon)$ with $0 < \varepsilon \leq \frac{1}{2}$. The words with counts $(3,1)$ are 1112, 2111, 1121 and 1211, and those with counts $(2,2)$ are 1122, 2211, 1221, 2112, 1212 and 2121. Counting changes gives $P_4^\mu(3,1) = p^3(u + 2\mu_1 u^2)$ and $P_4^\mu(2,2) = p^3(u + u^2 + u^3)$, so

$$P_4^\mu(3,1) - P_4^\mu(2,2) = p^3 u^2 (2\varepsilon - u).$$

So the unbalanced bin is more likely when $u < 2\varepsilon$, and for $\mu = \delta_1$ at every $\frac{1}{2} < p < 1$.

This example only shows that balancing can fail. For every μ that is not constant on F , the next-term expansion that Paper C states without the details [3] predicts more. For all large N and $cL \leq T \leq CL$, it predicts that the largest bin on F beats the balanced bins by a factor $1 + \frac{1}{4\mu(F)^2T} \sum_{a \in F} (\mu_a - \mu(F)/k)^2 + o(1/T)$. So balancing should fail there. We do not prove this.

Now we compare all faces at once. Each face gets a line in the window coordinate t , and the highest line wins.

Remark 4.5 (global modes with a start). This remark uses a result of Paper C. Order the start weights as $\mu_{(1)} \geq \dots \geq \mu_{(d)}$, put $M_k = \mu_{(1)} + \dots + \mu_{(k)}$ and $T = L - \frac{1}{2} \log L + t$, and define the lines

$$\ell_1^\mu(t) = \log \mu_{(1)}, \quad \ell_k^\mu(t) = \log \frac{M_k}{k} + (k-1)(t - a_k) \quad (2 \leq k \leq d).$$

By Theorem 4.3(3), a balanced bin n of a face S with $k \geq 2$ states has $\log(P_N^\mu(n)/p^{N-1}) = \log \frac{\mu(S)}{k} + (k-1)(t - a_k) + o(1)$, and a vertex i has $\log \mu_i$. Paper C proves, from its general theorems on face maxima and tie windows with the constants K_F above, that the best bin of each face has the same limit and that the lines decide the modes. So if $\max_k \ell_k^\mu(t)$ is attained only at $k = k^*$, then for this t and all large N every global mode of P_N^μ has exactly k^* positive coordinates, and its support S has $\mu(S) = M_{k^*}$.

The winning size is nondecreasing in t . Indeed, if k_1 wins at t_1 and k_2 at $t_2 > t_1$, adding the 2 comparisons gives $(k_2 - k_1)(t_2 - t_1) \geq 0$. Put $t^* = a_d + \frac{1}{d-1} \log(d\mu_{(1)})$, where ℓ_1^μ and ℓ_d^μ cross. Their maximum is convex with its only kink at t^* , and the other lines have slopes strictly between 0 and $d-1$. So some size $2 \leq k \leq d-1$ is the unique winner on an open interval of t if and only if $\ell_k^\mu(t^*) > \ell_1^\mu(t^*)$ for some such k . For the uniform start $\ell_k^\mu(t^*) - \ell_1^\mu(t^*) = (k-1)(a_d - a_k) < 0$, which agrees with Corollary 3.9.

Remark 4.6 (the start decides which face wins). Let $d = 3$ and $\mu = (\frac{1}{2}, \frac{1}{2}, 0)$, and put $t_{\text{ef}} = \log \frac{3}{2} + 2a_3 - a_2 = \frac{1}{2} \log \frac{32\pi}{243}$. In the terms of Remark 4.5, the lines are $\log \frac{1}{2}$ for the vertices 1 and 2, $\log \frac{1}{2} + t - a_2$ for the edge $\{1, 2\}$, $\log \frac{1}{4} + t - a_2$ for the edges through 3, and $\log \frac{1}{3} + 2(t - a_3)$ for the full face. The edge $\{1, 2\}$ has the highest line exactly when $a_2 < t < t_{\text{ef}}$. So for every fixed t in this window and all large N , the global modes are the balanced bins of the edge $\{1, 2\}$, since Theorem 3.8 applies on the edge, where μ is constant. For fixed $t < a_2$ the modes are $(N, 0, 0)$ and $(0, N, 0)$, and for fixed $t > t_{\text{ef}}$ they have 3 positive counts. The window is $(-0.4673558, -0.4412978)$, of width $\log \frac{16}{9\sqrt{3}} = 0.0260580$, while with the uniform start no proper face wins (Corollary 3.9). Paper C proves this window from its general tie-window theorem. Dynamic programs over all bins at $N = 90, 300$ and 1002 confirm these modes at finite N (numerically verified, see Section 7).

Remark 4.7 (the window at finite N). The balanced bins of the edge cross the vertices exactly at Paper A's root u_N , for every $N \geq 2$, since μ is constant on $\{1, 2\}$. Indeed $P_N^\mu(n_1, n_2, 0)/P_N^\mu(N, 0, 0) = S_{(n_1, n_2)}(u)$, which is the polynomial of Paper A [2] at $u = r/p$. This crossing is at $T = Nu_N/(1 + 2u_N)$, which differs from Paper A's $z_N = Nu_N/(1 + u_N)$ by $O(L^2/N)$. Near $t = a_2$ the lines of the full face and of the edges through 3 stay below those of the vertices, so for large N this crossing is the lower end of the window. For the 2 ends at finite N we use $z = Nu = T/p$. A heuristic calculation, which we do not prove, gives

$$z - L + \frac{1}{2} \log z = a_2 + \frac{1}{8z} + \frac{z^2}{2N} + \dots, \quad z - L + \frac{1}{2} \log z = t_{\text{ef}} + \frac{1}{12z} + \frac{3z^2}{2N} + \dots,$$

where only $a_2 + \frac{1}{8z}$ is proved, by Theorem 3.2. The terms z^2/N come from a discrete term $-\frac{(k-1)^2}{2} \frac{z^2}{N}$ in the logarithm of (6), and the upper end also uses a gain $\frac{1}{24z}$ of the best full-face bin over the balanced one. Without the terms z^2/N the width is $0.0260580 - 0.0546957/L + \dots$.

Exact computations give the widths 0.0843, 0.0324, 0.0223 and 0.0214 at $N = 300, 3000, 30000$ and 300000, and the prediction gives 0.0802, 0.0321, 0.0226 and 0.0217 (numerically verified, see Section 7). Below N of about 10^4 the terms z^2/N are larger than the terms $1/L$.

5 A weak Potts field as a tilt

In this section the start is uniform again, and we tilt the board slightly. Give each visit to state i the extra weight $e^{h_i/N}$, for a fixed vector $h = (h_1, \dots, h_d) \in \mathbb{R}^d$. Over N steps these weights multiply to $e^{h \cdot n/N}$, a factor of order 1. Again order 1 is exactly the scale of the constants a_k , so the field can reorder the faces. Formally we study the *tilted occupation law*

$$\tilde{P}_{N,h}(n) = \frac{e^{h \cdot n/N} P_N(n)}{\sum_{m_1 + \dots + m_d = N} e^{h \cdot m/N} P_N(m)}. \quad (9)$$

We use the window coordinate $T = L - \frac{1}{2} \log L + t$. The normalizing sum cancels whenever we compare 2 bins. When we speak of a modal phase we mean the support of the global modes, which need not be the face with most of the mass (Remark 5.8).

Proposition 5.1 (the field is a tilt). *Let $\Lambda_N(h) = \log \mathbb{E} \exp\langle h, K_N/N \rangle$.*

- (1) $\tilde{P}_{N,h}(n) = \exp(\langle h, n/N \rangle - \Lambda_N(h)) P_N(n)$. So $\tilde{P}_{N,h}$ is the exponential tilt of the law of the occupation rate K_N/N by the linear function $\langle h, \cdot \rangle$.
- (2) It is also the law of the counts when the law of the path is tilted by $\exp(\sum_{s=1}^N h_{X_s}/N)$. This is the Potts chain with free boundaries, coupling p/r per equal neighbor pair and field h/N at each site.
- (3) If $T \rightarrow \infty$, then $\Lambda_N(h) \rightarrow \bar{h} = \frac{1}{d} \sum_i h_i$.

Proof. Part (1) is (9) with its denominator written as $e^{\Lambda_N(h)}$. For (2), a word $x_1 \dots x_N$ has probability $\frac{1}{d} r^{N-1} (p/r)^{\sum_{s < N} \delta(x_s, x_{s+1})}$, and $\sum_s h_{x_s} = \langle h, n \rangle$ depends on the word only through its counts n . So tilting the words and then counting gives (1).

For (3), the chain is stationary with transition matrix $\sigma I + (1 - \sigma) \frac{1}{d} J$, whose m -th power is $\sigma^m I + (1 - \sigma^m) \frac{1}{d} J$. So $\mathbb{E} K_{N,a} = N/d$, and the indicators of state a at times s and $s + m$ have covariance $\sigma^m (\frac{1}{d} - \frac{1}{d^2})$. Summing over the 2 times gives, for $p < 1$,

$$\text{Var } K_{N,a} = \left(\frac{1}{d} - \frac{1}{d^2} \right) \left(N \frac{1 + \sigma}{1 - \sigma} - \frac{2\sigma(1 - \sigma^N)}{(1 - \sigma)^2} \right).$$

We bound this in 2 cases. Note that $1 - \sigma = dr$. If $1/d \leq p < 1$, then $0 \leq \sigma < 1$, the second term in the bracket is not positive, and $\text{Var}(K_{N,a}/N) \leq 2/(d^2 T)$. If $p < 1/d$, then $-1 \leq \sigma < 0$, the bracket is at most $N + 4$, and $T \leq N$, so $\text{Var}(K_{N,a}/N) \leq 5/T$. In both cases K_N/N tends to the uniform vector in probability as $T \rightarrow \infty$. Since $\exp\langle h, K_N/N \rangle$ is bounded, bounded convergence gives the limit. $\square$

Note the scale. By (1), $|\log \tilde{P}_{N,h}(n) - \log P_N(n)| \leq 2 \max_i |h_i|$ for every n . In the window this is of order 1, the same order as the constants a_k that break the tie. At first order, with $T = \tau L$, it is invisible, so a field of total size $o(\log N)$ does not change the first-order face selection of Paper C [3]. The Potts chain in a field is exactly solvable in the thermodynamic limit [16], but here N is finite and the field has size h_i/N . For a support S with $|S| \geq 2$ put $\bar{h}_S = \frac{1}{|S|} \sum_{i \in S} h_i$.

Theorem 5.2 (weak-field modal phase diagram). Put $T = L - \frac{1}{2} \log L + t$ and define the face scores

$$\ell_S(t, h) = \begin{cases} h_i, & S = \{i\}, \\ (|S| - 1)(t - a_{|S|}) + \bar{h}_S, & |S| \geq 2. \end{cases}$$

The largest bin weight, taken relative to the untilted vertex and before the normalization in (9), satisfies, uniformly for (t, h) in compact sets,

$$\log \max_n \left\{ e^{h \cdot n/N} \frac{P_N(n)}{V_N} \right\} \longrightarrow \max_{\emptyset \neq S \subseteq \{1, \dots, d\}} \ell_S(t, h). \quad (10)$$

For fixed (t, h) and all large N , every global mode has a support that maximizes the limiting score. If the maximizing support is unique, all global modes have exactly that support. On a selected support S with $|S| \geq 2$, their locations satisfy

$$\max_{i \in S} \left| \frac{n_i}{N} - \frac{1}{|S|} \right| = o((\log N)^{-1/2}).$$

This does not identify the maximizing vectors exactly, and equal limiting scores do not imply that both supports hold modes at finite N .

If the fields are sorted as $h_1 \geq \dots \geq h_d$, the maximum of the scores is the upper envelope of the d lines

$$\ell_1(t) = h_1, \quad \ell_k(t) = (k-1)(t - a_k) + \frac{1}{k} \sum_{i=1}^k h_i, \quad 2 \leq k \leq d. \quad (11)$$

Away from ties, the winning support size is nondecreasing in t . If the fields are strictly ordered, the winning supports are the nested sets $\{1, \dots, k\}$. Every support size $k \in \{2, \dots, d-1\}$ is the unique winning size on an open interval of t for suitable fixed fields. So proper-face phases persist under a field of order $1/N$ per step.

In words, the field shifts the line of each face by the field averaged over the face. Here is why. On a support S the ratio of a bin to a vertex has a Gaussian profile around the center of the face, of width $1/\sqrt{L}$ (Lemma B.1). The tilt multiplies it by $e^{h \cdot n/N}$, which is $e^{\bar{h}_S + o(1)}$ near the center. So the best bin on S scores $\ell_S(t, h)$, and taking the k largest fields gives the lines. The proof is in Appendix B.

Remark 5.3 (how many bins beat a vertex). The same profile counts the bins that beat a vertex. Put $h_* = \max_i h_i$ and, for a support S with $|S| = k \geq 2$, $B_S(t, h) = \ell_S(t, h) - h_*$. Let $M_{N,S}(t, h)$ be the number of bins with support exactly S whose tilted probability is strictly greater than that of every vertex. Then, uniformly for (t, h) in compact sets,

$$\frac{L^{(k-1)/2}}{N^{k-1}} M_{N,S}(t, h) \longrightarrow \frac{|\mathbb{B}_{k-1}|}{\sqrt{k}} \left(\frac{4[B_S(t, h)]_+}{k^2} \right)^{(k-1)/2},$$

where $|\mathbb{B}_{k-1}|$ is the volume of the unit ball in $\mathbb{R}^{k-1}$ and $[x]_+ = \max(x, 0)$ (proof omitted). In the coordinates $y = \sqrt{L}(n/N - \frac{1}{k}\mathbf{1})$ these bins fill a ball of squared radius $4B_S/k^2$, and $\sqrt{k}(\sqrt{L}/N)^{k-1}$ is the covolume of the lattice of the points y . For $k = 2$ the constant is $\sqrt{2[t - a_2]_+}$, which agrees with the binary window of Paper A once the different centering is taken into account. The profile and the lattice volumes are checked numerically (see Section 7), and the covolume $\sqrt{k}$ is checked in Lean.

On K_d the quasi-ergodic law of a face, the product of its left and right Perron vectors in Paper C, is uniform on the face, and $\bar{h}_S$ is the average of h under it. For a general generator Q we have $P_N \text{diag}(e^{h/N}) = I + \frac{T}{N}(Q + \text{diag}(h)/T) + O(T/N^2)$, so the field acts as a potential h/T added to Q . First-order perturbation of the Perron root then lowers $T\lambda_F$ by $\langle h, \alpha_F^* \rangle + O(1/T)$, where α_F^* is the quasi-ergodic law. So we expect each face line to gain $\langle h, \alpha_F^* \rangle$ in general, but we have not written this out.

Example 5.4 (a persistent triangular edge phase). For $d = 3$ and $h = (0, 0, -H)$ the 3 competing scores are 0, $t - a_2$ and $2(t - a_3) - H/3$. If $H > 6(a_2 - a_3)$, the edge $\{1, 2\}$ wins strictly for

$$a_2 < t < H/3 + 2a_3 - a_2,$$

and for every fixed t in this interval all global modes of the tilted law lie on that edge for large N . Without a field no proper face holds a mode in the window (Corollary 3.9).

Remark 5.5 (start versus field). Compare a start μ with full support and no field (Remark 4.5) with the uniform start and the field $h_i = \log(d\mu_i)$ (Theorem 5.2). We add $\log d$ to the start lines, which does not change the modes. Then a vertex i scores $\log(d\mu_i)$ in both. A support S with $k \geq 2$ states scores $(k - 1)(t - a_k)$ plus the logarithm of the arithmetic mean of $d\mu$ on S under the start, and plus the logarithm of the geometric mean under the field. The arithmetic mean is at least the geometric mean, with equality only when μ is constant on S . So a start favors mixed faces more than the field with the same vertex scores. For example, in Remark 4.6 state 3 has start weight 0, which as a field would mean $H = \infty$, and yet the edge window has width only 0.026. We have not written out the lines for a start and a field together.

One field can make every support size win in turn, from 1 state to all d .

Corollary 5.6 (one field realizes the full face hierarchy). *Fix $d \geq 2$ and $H > 3/4$, and let $h_i = -H(i - 1)^2$ for $1 \leq i \leq d$. In particular $h_i = -(i - 1)^2$ works in every fixed dimension. Put*

$$c_k = (k + \frac{1}{2}) \log k - \frac{k - 1}{2} \log(4\pi), \quad t_k = c_k - c_{k+1} + \frac{H(4k - 1)}{6} \quad (1 \leq k < d),$$

and $t_0 = -\infty$, $t_d = +\infty$. Then $t_1 < \dots < t_{d-1}$, and for every $k \in \{1, \dots, d\}$ the support $\{1, \dots, k\}$ is the unique winning support on the nonempty interval $t_{k-1} < t < t_k$. So for each fixed t in this interval and all large N , every global mode of the tilted law has exactly those k positive coordinates.

Here $c_1 = 0$ and $c_k = \log D_k$ for $k \geq 2$. Figure 3 shows the lines for $d = 5$, and the proof is in Appendix B.

Remark 5.7 (sticky priors). Our chain is Paper C's sticky chain with the uniform kernel. Paper C normalizes that chain by $Q = (J - dI)/(d - 1)$, so its T and τ are $d - 1$ times ours, and the tie at our $T = L$ sits at its $\tau = d - 1$. The uniform kernel meets Paper C's direct-jump criterion with equality for every face. For $d \geq 3$ and a symmetric kernel with all off-diagonal rates positive, Paper C shows that only the uniform kernel meets this criterion, so only it ties on every face at first order. The tie then resolves as a direct jump (Corollary 3.9), while a field (Corollary 5.6) or a start that is not uniform (Remark 4.6) puts intermediate faces inside the window, of width $O(1/N)$ in p . If the path is observed through a likelihood with log-likelihood $\langle h, n \rangle / N$, then $\tilde{P}_{N,h}$ is the posterior law of the counts. Kernels that are not uniform act at first order in Paper C, on the scale L in T , and the field and the start only at order 1, so they are not interchangeable.

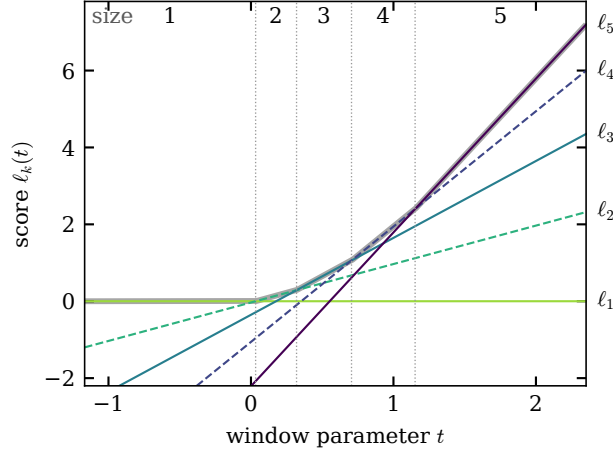


Figure 3: The lines (11) for $d = 5$ and $h_i = -(i-1)^2$. The grey band is their upper envelope. The dotted lines are $t_1 < \dots < t_4$, and the number at the top of each interval is the size k of the winning support $\{1, \dots, k\}$.

Remark 5.8 (mass versus mode). From each state of a face F with k states the chain stays in F with probability $1 - (d-k)r$, so $\mathbb{P}(\text{supp } K_N \subseteq F) = \mu(F)(1 - (d-k)r)^{N-1}$. Every bin without full support lies on a face with $d-1$ states. So the bins with full support have mass at least $1 - \sum_{|F|=d-1} \mu(F)(1-r)^{N-1} = 1 - O(e^{-T})$, while for the uniform start, $t < a_d$ and large N the modes are single states (Corollary 3.9).

6 Near the boundary of the simplex

So far every positive coordinate was of order N (Theorem 3.3 needs each to be at least εN). What about bins with some small coordinates, near an edge or a vertex of the triangle? In this section we drop that assumption. We first give an upper bound that holds for every positive count vector (Theorem 6.2). Then we find the scale at which a small coordinate starts to matter. It is N/L^2 when at least 2 coordinates are of order N (Remark 6.3). Near a vertex the picture changes: the crossing moves to odds of order $N^{-1/2}$, and the small coordinates alone decide it (Subsection 6.1).

The start is uniform, but note that $S_n(u)$ does not involve the start. Proposition 2.2 gives $P_N^\mu(n) \leq (\max_i \mu_i) p^{N-1} S_n(u)$ for every start μ , so every upper bound below also holds for a general start. The basic tool compares the Laguerre-type polynomials of the ratio form with the Bessel series Φ .

Lemma 6.1 (a uniform Laguerre–Bessel inequality). *For every integer $a \geq 1$ and real $w > 0$,*

$$0 \leq 1 - \frac{B_a(w)}{w\Phi(aw)} \leq \frac{w}{2}, \quad B_a(w) = \sum_{m=1}^a \binom{a-1}{m-1} \frac{w^m}{m!}. \quad (12)$$

The proof, in Appendix C, compares coefficients with Lemma A.2.

Theorem 6.2 (a uniform Bessel majorant on every simplex stratum). *Let $n_1, \dots, n_k$ be arbitrary positive integers with sum N , and let $0 < u < 1$, $v = u/(1-u)$. Define*

$$\mathcal{Q}_n(u) = (1-u)^{N-1} v^{k-1} \int_0^\infty e^{-t} t^k \prod_{i=1}^k \Phi(n_i v t) dt.$$

Then, with no lower bound on any n_i/N ,

$$0 \leq 1 - \frac{S_n(u)}{\mathcal{Q}_n(u)} \leq k(k+1)v + \frac{k}{2}v^2 \left(\sum_i \sqrt{n_i} \right)^2 \leq k(k+1)v + \frac{k^2}{2}Nv^2. \quad (13)$$

Thus this approximation has relative error $O_k(L^2/N)$ whenever $u = O(L/N)$, uniformly over every positive composition of N .

In words, $\mathcal{Q}_n$ is an upper bound for every bin, and it is sharp up to a relative error $O(L^2/N)$ in the range of the crossings. Here is the idea of the proof, which is in Appendix C. We apply Lemma 6.1 to each factor of an integral form of S_n . Then the relative error is at most $kv/2$ times the mean of t under the probability density proportional to $e^{-t}t^k \prod_i \Phi(n_i vt)$, and an integration by parts bounds this mean by $2(k+1) + v(\sum_i \sqrt{n_i})^2$.

Remark 6.3 (the scale $N/(\log N)^2$). Here is the heuristic. A small coordinate a adds its own runs. Each run needs about 2 switches, which gives a factor u^2 , and it can go in about N places. So what matters is aNu^2 . At the crossings of Section 3 the value $Nu = T/p$ is of order L , and then aNu^2 is of order aL^2/N . So when at least 2 coordinates are of order N , a rare count starts to matter once it has size $\rho N/L^2$ with ρ fixed.

Appendix C proves this. Take $k_1 \geq 2$ large coordinates with fractions β_i among themselves and ℓ rare counts $a_j = \rho_j N/L^2$. With $z = Nu$, Theorem C.1 gives $S_n(z/N)$ up to a factor $1 + O(L^{-2} + L^2/N)$. Each rare count contributes a factor $\Phi(A^2 a_j z^2/N)$, where $A = \sum_i \sqrt{\beta_i}$, and the complete correction of relative order $1/L$ is a factor $e^{\mathcal{E}(z)}$ with 3 terms. The theorem also gives the crossing as the root of an implicit equation, and as an explicit expansion in L , which converges slowly. Rare counts with $a_j = o(N/L^2)$ change that expansion only by $o(1)$ (Corollary C.3), but their number ℓ changes its leading and logarithmic coefficients. With only one large coordinate the formula would have $A = 1$, so the growth factor $e^{(A^2-1)z}$ is gone. Then the crossing moves to u of order $b^{-1/2}$, where b is the large count, and the scale changes. This is the case of Subsection 6.1.

6.1 Near a vertex

Now let one count b be large and let the other positive counts $a_1, \dots, a_\ell$ be small compared with b , with $\ell \geq 1$ fixed. In this subsection and in Appendix D we put $|a| = a_1 + \dots + a_\ell$. These rare counts are not the constants a_k of Section 3, which do not appear here. With B_a as in Lemma 6.1, put

$$H(y) = \prod_{i=1}^{\ell} B_{a_i}(y), \quad \kappa(y) = \frac{yH'(y)}{H(y)}.$$

The polynomial H has positive coefficients, $H(0) = 0$ and $H(1) \geq 1$, so $H(y_a) = 1$ has exactly one positive root y_a , and $y_a \leq 1$. Write $S_{(b,a)}$ for S_n with $n = (b, a_1, \dots, a_\ell)$. It is a polynomial in u with nonnegative coefficients, zero constant term and a positive coefficient at u^ℓ , so $S_{(b,a)}(u) = 1$ has exactly one positive root $u_{b,a}$.

Each rare run again needs about 2 switches and has about b places, so the rare counts enter through bu^2 . For fixed rare counts the bounds of Lemma D.2 in Appendix D give $S_{(b,a)}(\sqrt{y/b}) \rightarrow H(y)$ as $b \rightarrow \infty$. So the crossing is near $\sqrt{y_a/b}$, and Nu is of order $\sqrt{N}$ instead of L . The next theorem shows that this leading term stays correct uniformly as long as $|a| = o(b)$.

Theorem 6.4 (all growing rare counts near a vertex). *Fix $\ell \geq 1$. Uniformly over all positive rare counts with $|a| = o(b)$, the unique positive crossing odds satisfy*

$$u_{b,a} = \sqrt{\frac{y_a}{b}} \left(1 + O_\ell \left(\sqrt{\frac{|a|+1}{b}} \right) \right), \quad \prod_i B_{a_i}(y_a) = 1. \quad (14)$$

Explicitly, put

$$c_\ell = \ell + \sqrt{\ell} + 1, \quad \varepsilon = \sqrt{\frac{|a|+1}{b}}, \quad \delta = 16c_\ell\varepsilon.$$

If $\delta \leq 1/2$, then

$$(1 - \delta)\sqrt{y_a/b} \leq u_{b,a} \leq (1 + \delta)\sqrt{y_a/b}. \quad (15)$$

The constants are not optimized.

In words, near a vertex the crossing is decided by the polynomial H of the rare counts alone, evaluated at bu^2 , as long as the rare counts have total $o(b)$. The bracket (15) holds for all b and all positive rare counts with $\delta \leq 1/2$, and it implies (14).

The proof is in Appendix D. It rests on 2 facts. First, $\log H$ is concave, and for $0 < y \leq 1$ its elasticity $\kappa(y)$ lies between $\max\{\ell, \frac{1}{2}\sqrt{|a|y}\}$ and $\ell + \sqrt{\ell|a|y}$ (Lemma D.1). Second, $S_{(b,a)}(u) = (1-u)^{|a|}\mathbb{E}[H(v\mathcal{T})]$ exactly, where $v = u/(1-u)$ and $\mathcal{T}$ is a gamma variable plus a compound binomial sum with mean $2 + (b-1)u$ (Lemma D.2). Jensen's inequality then gives $S_{(b,a)}(u) \geq (1-u)^{|a|}H(bu^2)$, which exceeds 1 at $u = (1+\delta)\sqrt{y_a/b}$. The tangent line of $\log H$ at bu^2 and the moment generating function of $\mathcal{T}$ give an upper bound that is below 1 at $u = (1-\delta)\sqrt{y_a/b}$.

We have also found the next term of $u_{b,a}$ for fixed rare counts, which for one rare count is the 2-state fixed-edge expansion of Paper A, the growth $u_{b,a} \sim \ell \log |a| / (2A\sqrt{|a|b})$ with $A = \sum_i \sqrt{a_i/|a|}$ when $|a| \rightarrow \infty$ with proportions $a_i/|a|$ bounded away from 0, and an example with $|a| = o(b)$ where $S_{(b,a)}(u) \sim H(bu^2)$ fails although the root is still correct (proofs omitted, checked numerically, see Section 7).

7 Verification and open questions

No proof in this paper relies on computation. All 4 papers on Problem 131 share one verification supplement, `SUPPLEMENT.md` in the Problem 131 folder of <https://github.com/apemm/Kagey-Problems>, and `python -B verify_all.py --paper b` in that folder runs the checks for this paper (`--full` adds a run of about 3 minutes). The scripts use only the Python standard library. They compare the exact law with dynamic programs and with enumeration of all words, and they check the crossings, the constants and their tree forms. The statements that we mark as numerically verified or with the proof omitted are checked as follows, each in 2 ways where possible.

- Remark 4.2: `verify_general_start.py` computes the start weights from the run form and from the refresh form in double precision, at $N = 2000$ with 2 states, against the limit.
- Remark 4.6: dynamic programs over all bins in double precision, at $N = 90$ in `verify_general_start.py` and at $N = 300$ and 1002 in `verify_B_tie_repair.py`.
- Remark 4.7: the 2 ends of the window by bisection in double precision, the lower end from the run form and from Paper A's polynomial, at $N = 300$ in `verify_B_tie_repair.py` and at $N \geq 3000$ in `verify_general_start.py`.
- Remark 5.3: `verify_frontier_modes.py` evaluates the refresh series in 65-digit decimal arithmetic and counts lattice points against the volume formula.

- The last paragraph of Subsection 6.1: `verify_frontier_boundary.py` and `verify_vertex_completion.py` check exact identities in rational arithmetic against word enumeration, and evaluate positive sums in 70-digit or 100-digit arithmetic at 2 series cutoffs.
- The numbers in the remark after Theorem C.1: `verify_crossover_factor.py`, as described at the end of that remark.

None of these checks is interval-certified. Before the newer computations we wrote the predictions in ledgers, which keep the 2 predictions that failed. Many exact identities and finite bounds are also proved in Lean 4, in the folder `lean/PaperB`: among them the exact law for the uniform start, balancing, the order of the constants a_k , the arithmetic of Corollary 5.6 and the bracket near a vertex. The asymptotic results are not in Lean.

Several questions remain open. We have not proved the discrete term $-\frac{(k-1)^2}{2}z^2/N$ or the second-order ends of the window of Remark 4.7. We have not proved that balancing fails at all large N for every start that is not constant on the face. Corollary 3.9 has no effective N_0 , and by Example 3.10 it does not extend to every N . We say nothing about the modes when T is much larger than L . Near a vertex we know the leading crossing uniformly for rare counts of total $o(b)$, but not the next term uniformly in that range, and not with a start that is not uniform. We have not treated the crossover with a start that is not constant on the support, in particular one that sits only on the rare coordinates, or a number of states that grows with N . Switching that is not uniform is the subject of Paper C.

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A Proofs for Sections 2 and 3

A.1 Facts about Bessel series

The weights $x^m/(m!(m+1)!)$ of the series Φ define a probability law on the integers $m \geq 0$. Its moments control every Laplace expansion in this paper.

Lemma A.1 (the Bessel law). *For $x \geq 0$ let J be a random variable with $\mathbb{P}(J = m) = x^m/[m!(m+1)!\Phi(x)]$ for $m \geq 0$ (so $J = 0$ when $x = 0$), and put $\psi(x) = \mathbb{E}J$. Then*

$$(i) \quad \mathbb{E}[J(J+1)] = x,$$

$$(ii) \quad \mathbb{E}[J(J-1)J(J+1)] = x^2,$$

$$(iii) \quad \psi(x) = x\Phi'(x)/\Phi(x), \text{ and } \psi(x) = \sqrt{x} I_2(2\sqrt{x})/I_1(2\sqrt{x}) \text{ for } x > 0,$$

$$(iv) \quad x\psi'(x) = \text{Var } J, \text{ so } 0 \leq \psi'(x) \leq 1,$$

$$(v) \quad 0 \leq \psi(x) \leq \min(x/2, \sqrt{x}),$$

$$(vi) \quad 0 \leq (\log \Phi)'(x) = \psi(x)/x \leq \frac{1}{2} \text{ for } x > 0.$$

Proof. (i) We have $\sum_{m \geq 1} m(m+1)x^m/[m!(m+1)!] = \sum_{m \geq 1} x^m/[(m-1)!m!] = x\Phi(x)$. (ii) For $m \geq 2$, $m(m-1)/m! = 1/(m-2)!$ and $m(m+1)/(m+1)! = 1/(m-1)!$, so the sum is $\sum_{m \geq 2} x^m/[(m-2)!(m-1)!] = x^2\Phi(x)$. (iii) The first form is the definition of the mean. Termwise, $\Phi'(x) = \sum_{i \geq 0} x^i/[i!(i+2)!] = I_2(2\sqrt{x})/x$ by the series of I_2 [18, Eq. 10.25.2], which gives the

second. (iv) Since $\log \mathbb{P}(J = m) = m \log x - \log \Phi(x)$ plus a term free of x , we have $x \frac{d}{dx} \log \mathbb{P}(J = m) = m - \psi(x)$. So $x\psi'(x) = \mathbb{E}[J(J - \psi)] = \text{Var } J$, and $\text{Var } J \leq \mathbb{E}J^2 \leq \mathbb{E}[J(J + 1)] = x$. (v) For integers $J \geq 0$ we have $J \leq J(J + 1)/2$, so $\psi \leq x/2$ by (i). Also $\psi^2 \leq \mathbb{E}J^2 \leq x$. (vi) follows from (iii) and (v). $\square$

Lemma A.2 (clipped product deficit). *For any finite list of nonnegative real numbers $t_1, \dots, t_m$, with $(x)_+ = \max(x, 0)$,*

$$0 \leq 1 - \prod_{j=1}^m (1 - t_j)_+ \leq \sum_{j=1}^m t_j.$$

Proof. If some $t_j \geq 1$, the product is zero and the sum is at least one. Otherwise all factors lie in $[0, 1]$, and induction on m follows from $1 - (1 - a)(1 - b) = a + b - ab \leq a + b$ for $a, b \in [0, 1]$. The empty product gives equality. $\square$

Lemma A.3 (Gaussian averages of the factors Φ). *Fix $q \geq 0$, an integer $\ell \geq 0$ and $X \geq 0$. Let $w_0 \geq 1$ and $c_1, \dots, c_\ell \geq 0$ with $\xi_j = c_j w_0^2 \leq X$, and put $F(w) = w^q \prod_{j=1}^\ell \Phi(c_j w^2)$. Let $J_1, \dots, J_\ell$ be independent, with J_j distributed as J in Lemma A.1 with $x = \xi_j$. Put $|J| = J_1 + \dots + J_\ell$ and $\mu_q = \mathbb{E}[(q + 2|J|)(q + 2|J| - 1)]$. Then*

$$\frac{1}{\sqrt{\pi}} \int_{-w_0/2}^{w_0/2} e^{-y^2} F(w_0 + y) dy = F(w_0) \left(1 + \frac{\mu_q}{4w_0^2} + O(w_0^{-4}) \right),$$

and $F(w_0 + y) = O(F(w_0))$ for $|y| \leq w_0/2$. The constants depend only on q, ℓ and X . Moreover

$$\mu_q = q^2 - q + 4 \sum_j \xi_j + (4q - 6) \sum_j \psi(\xi_j) + 8 \sum_{i < j} \psi(\xi_i) \psi(\xi_j).$$

Proof. Step 1: derivatives as moments. Expanding each Φ gives $F(w) = \sum_m \kappa_m w^{q+2|m|}$, where $m = (m_1, \dots, m_\ell)$ runs over vectors of nonnegative integers, $|m| = \sum_j m_j$ and $\kappa_m = \prod_j c_j^{m_j} / [m_j! (m_j + 1)!] \geq 0$. The weights $\kappa_m w_0^{q+2|m|} / F(w_0)$ are the joint law of $(J_1, \dots, J_\ell)$. Hence $F^{(i)}(w_0) = F(w_0) \mathbb{E}[(q + 2|J|)_i] w_0^{-i}$, where $(y)_i = y(y - 1) \dots (y - i + 1)$. In particular $F''(w_0) = \mu_q w_0^{-2} F(w_0)$.

Step 2: the fourth derivative. Let $w = w_0 + y$ with $|y| \leq w_0/2$. For every $\eta \geq 0$ we have $w^{\eta-4} \leq 16(3/2)^\eta w_0^{\eta-4}$ (consider $\eta \geq 4$ and $\eta < 4$ separately) and $|(\eta)_4| \leq \eta^4 + 1$. Applying this with $\eta = q + 2|m|$ termwise gives

$$|F''''(w)| \leq 16 \left(\frac{3}{2} \right)^q w_0^{q-4} \prod_j \Phi\left(\frac{9\xi_j}{4}\right) \tilde{\mathbb{E}}[(q + 2|\tilde{J}|)^4 + 1],$$

where the $\tilde{J}_j$ are independent with the parameters $9\xi_j/4$ in place of ξ_j . By Lemma A.1(i) and (ii), $\mathbb{E}J^4 = x^2 + \mathbb{E}J^2 \leq x^2 + x$. Hence $\tilde{\mathbb{E}}[(q + 2|\tilde{J}|)^4]$ is bounded in terms of q, ℓ and X . Since $\Phi \geq 1$ we have $F(w_0) \geq w_0^q$, so $F''''(w) = O(w_0^{-4} F(w_0))$. The same argument without derivatives bounds $F(w_0 + y)$.

Step 3: Taylor expansion. Expand $F(w_0 + y)$ to third order in y with the Lagrange remainder. The odd terms integrate to zero over the symmetric interval. The integrals of e^{-y^2} and $y^2 e^{-y^2}$ over $|y| \leq w_0/2$ differ from $\sqrt{\pi}$ and $\sqrt{\pi}/2$ by $O(w_0 e^{-w_0^2/4})$, the number μ_q is bounded, and the remainder term is $O(w_0^{-4} F(w_0))$. This proves the expansion.

Step 4: the formula for μ_q . Put $\psi_j = \psi(\xi_j) = \mathbb{E}J_j$. Then $\mathbb{E}J_j^2 = \xi_j - \psi_j$ by Lemma A.1(i), and independence gives $\mathbb{E}[(q + 2|J|)^2] = q^2 + 4q \sum_j \psi_j + 4 \sum_j (\xi_j - \psi_j) + 8 \sum_{i < j} \psi_i \psi_j$. Subtracting $\mathbb{E}[q + 2|J|] = q + 2 \sum_j \psi_j$ proves it. $\square$

A.2 Proofs for Section 3

Proof of the exact assertions in Theorem 3.2. Let n be a balanced vector with k positive coordinates, so that $C_{N,k}/V_N = S_n(u)$ with $u = (1-p)/((d-1)p)$.

Step 1: one crossing. To count $W_n(j)$, choose a sequence of $j+1$ run colors in which successive colors differ. If color i appears m_i times, its run lengths can be chosen in $\binom{n_i-1}{m_i-1}$ ways. So every coefficient of S_n is nonnegative, and the first nonzero one is $W_n(k-1) = k!$, from the $k!$ orders of k single runs. Hence S_n increases strictly from 0 to ∞ on $u > 0$ and equals 1 exactly once. As p increases over $(0, 1)$, u decreases over $(0, \infty)$, which gives the unique $p_{N,k}$ and the 2 inequalities.

Step 2: monotonicity in N . Choose the balanced vectors for increasing N by adding 1 to one of the smallest coordinates. Each coefficient $W_n(j)$ is then nondecreasing. To see this, map a word to the word with an extra copy of the incremented color just before its first occurrence. This keeps the number of changes, and it is injective, since deleting the first letter of the first run of that color inverts it. The coefficient of u^k is

$$W_n(k) = \frac{k!(k-1)}{2} \sum_i (n_i - 1) = \frac{k!(k-1)}{2} (N - k).$$

Indeed, exactly one run color appears twice. For each choice of it there are $(k+1)!/2 - k! = k!(k-1)/2$ sequences of run colors in which successive colors differ (all orders, minus those with the 2 copies adjacent), and $n_i - 1$ ways to split its n_i visits into 2 runs. This coefficient increases strictly with N . So at every $u > 0$ the value $S_n(u)$ increases strictly with N , the root of $S_n(u) = 1$ decreases strictly with N , and $p = 1/(1 + (d-1)u)$ increases.

Step 3: the first crossing. If $N = k$, every coordinate is 1 and $S_n(u) = k!u^{k-1}$, which gives $p_{k,k}$. Since the root decreases with N , every root has $u \leq (k!)^{-1/(k-1)} < 1$, and $u < 1$ means $r < p$, that is, $p > 1/d$. $\square$

The asymptotic assertions rest on a Laplace expansion of the positive sum (3). It is the case with no rare counts of the crossover theorem in Appendix C.

Lemma A.4 (no rare counts). *Fix $k \geq 2$, $\varepsilon > 0$ and $0 < c < C$. Let n be a positive count vector with k coordinates and total N , put $\beta = n/N$, and assume that every $\beta_i \geq \varepsilon$. Put $A = \sum_i \sqrt{\beta_i}$, $g = A^2 - 1$ and $s = k - 1$, let $D(\beta)$ be as in Theorem 3.3, and put*

$$\gamma_0(\beta) = \frac{k(k+2)}{16A^2} - \frac{3}{16A} \sum_{i=1}^k \beta_i^{-1/2}.$$

Then, uniformly for these n and $cL \leq z \leq CL$,

$$S_n(z/N) = D(\beta) \frac{z^{s/2} e^{gz}}{N^s} e^{\gamma_0(\beta)/z} \left(1 + O\left(\frac{1}{L^2} + \frac{L^2}{N}\right) \right). \quad (16)$$

Proof. This is Theorem C.1 with $\ell = 0$ and $k_1 = k$. Then $M = N$, $\eta = s/2$, the constant $D_{k_1}(\beta)$ is $D(\beta)$, there are no factors Φ , and $\mathcal{E}(z) = \gamma_0/z$ with γ_0 as above, since $(2\eta+1)(2\eta+3) = k(k+2)$. The proof of that theorem uses only Theorem 6.2, the expansion of I_1 and Lemma A.3, and nothing from Section 3. $\square$

Proof of the asymptotic assertions in Theorems 3.1 and 3.2. Put $s = k - 1$ and let n be balanced, so that $\beta_i = n_i/N = 1/k + O(1/N)$.

Step 1: the balanced ratio (6). At the center $A^2 - 1 = s$, $D(\beta) = D_k$ and $\gamma_0 = (k+2)/16 - 3k/16 = -s/8$. On the plane $\sum_i \beta_i = 1$ the function A^2 has a critical point at the center, so

$A^2 - 1 = s + O(N^{-2})$, while $D(\beta)$ and $\gamma_0(\beta)$ change by $O(1/N)$. Since $z = O(L)$, Lemma A.4 with $\varepsilon = 1/(2k)$ and $e^{-s/(8z)} = 1 - s/(8z) + O(z^{-2})$ gives (6).

Step 2: parts (i) to (iii). Let $cL \leq T \leq CL$ and $u = r/p$, so that $z = T/p$ lies in $[cL, 2CL]$ for large N and $z - T = (d-1)Tz/N = O(L^2/N)$. Take logarithms in (6), with c and $2C$ in place of c and C , and replace z by T . This changes each term by $O(T^2/N)$, and $\log D_k = -sa_k$, so we get (i). For (ii), $\log V_N = (N-1)\log(1 - (d-1)T/N) - \log d = -(d-1)T - \log d + O(T^2/N)$. With $T = \tau L$ and (i) we get $\log C_{N,k} = -(d-1)\tau L + (k-1)(\tau-1)L + O(\log L)$, which is (ii), and the case $k = 1$ is the formula for $\log V_N$. For (iii), $T = L - \frac{1}{2}\log L + t$ gives $\Psi_N(T) = t + \frac{1}{2}\log(T/L) = t + O(\log L/L)$ uniformly on compact t sets, and the terms $1/(8T)$ and the error tend to 0.

Step 3: the crossing. Part (i) at $T = L/2$ and at $T = 2L$ shows that the ratio tends to 0 and to ∞ . Since $S_n(u)$ increases with u and u increases with T , the root lies in $[L/2, 2L]$ for large N , and (i) applies there with $c = \frac{1}{2}$ and $C = 2$. The same argument at $T = (1 \pm \varepsilon)L$ gives $z_{N,k}/L \rightarrow 1$. At the root the left side of (i) is 0, and dividing by s gives the implicit equation, since $T^2/N = o(L^{-2})$. For the explicit form put $z_* = L - \frac{1}{2}\log L + a_k + e_L/L$ with $e_L = \frac{1}{4}\log L - \frac{1}{2}a_k + \frac{1}{8}$. Since $\log z_* = \log L - \frac{\log L}{2L} + \frac{a_k}{L} + O(\frac{(\log L)^2}{L^2})$, we get $z_* + \frac{1}{2}\log z_* - 1/(8z_*) = L + a_k + O((\log L)^2/L^2)$. The derivative of $z + \frac{1}{2}\log z - 1/(8z)$ is at least 1 for $z > 0$, so $z_{N,k} - z_* = O((\log L)^2/L^2)$, which is (5).

Step 4: part (iv). We have $a_k = \frac{1}{2}\log(4\pi) - f(k)$ with $f(x) = (x + \frac{1}{2})\log x/(x-1)$. For $x \geq 2$ the sign of $f'(x)$ is the sign of $\Delta(x) = 2x^2 - x - 1 - 3x\log x$. Here $\Delta(2) = 5 - 6\log 2 > 0$, and $\Delta'(x) = 4x - 4 - 3\log x > 0$ for $x \geq 2$, since $\Delta'(2) > 0$ and $\Delta''(x) = 4 - 3/x > 0$. So f increases strictly and a_k decreases strictly. $\square$

Proof of Theorem 3.3. The polynomial S_n has nonnegative coefficients and lowest term $k!u^{k-1}$, so $S_n(u) = 1$ has a unique positive root. Lemma A.4 with $\beta = \alpha$ gives (7), since $\gamma_0(\alpha)$ is bounded when every $\alpha_i \geq \varepsilon$ and $e^{\gamma_0/z} = 1 + O(1/z)$.

For the root, note that $A^2 = 1 + 2\sum_{i < j} \sqrt{\alpha_i \alpha_j}$, so $g \in [2\varepsilon, s]$, and $D(\alpha)$ has positive lower and finite upper bounds depending on k and ε . Put $c_0 = \frac{1}{2}$ and $C_0 = s/\varepsilon$. By (7), $\log S_n(c_0L/N) \leq (gc_0 - s)L + O(\log L) \leq -\frac{s}{2}L + O(\log L) < 0$ and $\log S_n(C_0L/N) \geq (gC_0 - s)L + O(\log L) \geq sL + O(\log L) > 0$ for large N . So the root has $Nu_n \in [c_0L, C_0L]$, and taking logarithms in (7) there gives

$$gz + \frac{s}{2}\log z = sL - \log D(\alpha) + O_{k,\varepsilon}(z^{-1} + z^2/N), \quad z = Nu_n.$$

First $z = (s/g)L + O_{k,\varepsilon}(\log L)$. Then $\frac{s}{2}\log z = \frac{s}{2}\log L + \frac{s}{2}\log(s/g) + O(\log L/L)$, and solving for z gives (8). The derivative of the left side is at least $g \geq 2\varepsilon$, so the error is preserved on inversion. Finally, the functions of α in the formula have bounded derivatives on the set $\{\alpha_i \geq \varepsilon/2\}$. So moving α by $O(1/N)$ changes the expression by $O(L/N)$, which the error term absorbs. $\square$

Proof of Lemma 3.4. Note that $(\mathcal{L}x)_i = \sum_j c_{ij}(x_i - x_j)$ and $\mathcal{L}\mathbf{1} = 0$. Take θ with $\theta_1 = 0$ and put $\zeta = \text{diag}(\pi)f$. Then $\sum_i \zeta_i = 0$, $\mathcal{L}\chi = \zeta$ and the form is $2\zeta^\top \chi$. The vector $\tilde{\chi} = \chi - \chi_1\mathbf{1}$ also solves $\mathcal{L}\tilde{\chi} = \zeta$, with $\zeta^\top \tilde{\chi} = \zeta^\top \chi$ and $\tilde{\chi}_1 = 0$. So rows 2 to k give $\mathcal{L}_{(1)}\tilde{\chi}' = \zeta'$, where primes delete the first coordinate, and the form is $2\zeta'^\top \mathcal{L}_{(1)}^{-1}\zeta'$. As $\theta_1 = 0$, we have $\zeta' = (D' - \pi'\pi'^\top)\theta'$ with $D' = \text{diag}(\pi_2, \dots, \pi_k)$. Hence $\Sigma = 2(D' - \pi'\pi'^\top)\mathcal{L}_{(1)}^{-1}(D' - \pi'\pi'^\top)$. The matrix determinant lemma gives $\det(D' - \pi'\pi'^\top) = \prod_{i \geq 2} \pi_i (1 - \sum_{i \geq 2} \pi_i) = \prod_i \pi_i$, and the matrix-tree theorem [17] gives $\det \mathcal{L}_{(1)} = \text{tree}(c) > 0$. $\square$

Proof of Proposition 3.6. (1) For $i \neq j$ we have $\alpha_i G_{ij}^\alpha = h_i h_j$, which is symmetric, so α is reversible and hence stationary. Also $(Q_k h)_i = A - kh_i$, so $\theta_{\alpha,i} = k - A/h_i$, and $Q_k + \text{diag}(\theta_\alpha)$ has off-diagonal

entries 1 and diagonal entries $1 - A/h_i$. It is symmetric and kills $h > 0$, so 0 is its Perron root. Conjugating by $\text{diag}(h)$ gives the rates h_j/h_i . If G^α were a Doob transform of Q_k , then h would be an eigenvector of Q_k , so $A - kh_i = \lambda h_i$ for all i and α would be uniform. (2) is Lemma 3.5 with $x_i = h_i$. (3) By (2), $\sqrt{\text{tree}(\alpha)}/\prod_i \alpha_i = (\prod_i \alpha_i)^{-3/4} A^{k/2-1}$, which gives the first form. Lemma 3.4 with $\pi = \alpha$ gives $\det(2\pi\Sigma_\alpha) = (4\pi)^{k-1}(\prod_i \alpha_i)^2/\text{tree}(\alpha)$, which gives the second. (4) At the center $\mathcal{L}_{(1)} = I - J/k$ and $D' - \alpha'\alpha'^\top = (I - J/k)/k$ in the formula $\Sigma = 2(D' - \alpha'\alpha'^\top)\mathcal{L}_{(1)}^{-1}(D' - \alpha'\alpha'^\top)$ from the proof of Lemma 3.4. This gives Σ_k , and the rest is substitution. (5) is the same computation with k_1 and β . $\square$

Proof of Proposition 3.7(2) and (3). Let $u = z/N$ with $cL \leq z \leq CL$, and put $s = k - 1$.

Step 1: the mass. In (1) the term $j = k$ is $k(1 + (k-1)z/N)^{N-1} = k e^{(k-1)z + O(z^2/N)}$, and each term with $j < k$ is smaller by a factor $O(e^{-z})$. This gives the first expression.

Step 2: the Hessian. Write $\alpha' = (\alpha_2, \dots, \alpha_k)$ and $g(\alpha) = A^2 - 1$ as in Theorem 3.3. In full coordinates the Hessian of A^2 at the center is $(k/2)J - (k^2/2)I$. Pulling it back along $w = (-\sum_i v_i, v)$ gives $-\nabla^2 g = (k^2/2)(I + J)$ in the coordinates α' , and its inverse is $(2/k^2)(I - J/k) = \Sigma_k$.

Step 3: the bins near the center. Fix a small $\eta > 0$ and let K_η be the set of α with $|\alpha_i - 1/k| \leq \eta$ for all i . On K_η , Theorem 3.3 gives $S_n = D(\alpha)z^{s/2}e^{g(\alpha)z}N^{-s}(1 + O(z^{-1} + z^2/N))$ uniformly, with $\alpha = n/N$. The points α' form a lattice with cells of volume N^{-s} , and on each cell g changes by $O(1/N)$ and D by a factor $1 + O(1/N)$. So the sum over $n/N \in K_\eta$ is N^s times the integral of the same expression over K_η , times $1 + O(z/N)$, up to the cells near the boundary of K_η . There $g \leq k - 1 - c_\eta$ for some $c_\eta > 0$, so these cells are negligible. The function g has its strict maximum $k - 1$ at the center, and D is continuous there with $D = D_k$. So Laplace's method gives

$$\int_{K_\eta} D(\alpha)e^{g(\alpha)z} d\alpha' = D_k e^{(k-1)z} \left(\frac{2\pi}{z}\right)^{s/2} (\det \Sigma_k)^{1/2} (1 + o(1)).$$

Multiplied by $z^{s/2}$ this is the second expression, provided the bins off K_η are negligible.

Step 4: the other bins. For them, Theorem 6.2 and $\Phi(x) \leq e^{2\sqrt{x}}$ give, with $Y = Nv$ and $A_n = \sum_i \sqrt{n_i/N}$,

$$S_n(u) \leq (1-u)^{N-1} v^s \int_0^\infty e^{-t+2A_n\sqrt{Y}t} t^k dt \leq C' \frac{(1+z)^{2k}}{N^s} e^{(A_n^2-1)z + O(z^2/N)},$$

where we wrote $-t + 2A_n\sqrt{Y}t = A_n^2 Y - (\sqrt{t} - A_n\sqrt{Y})^2$. Here $A_n^2 \leq k - c'_\eta$ for some $c'_\eta > 0$, since A^2 is continuous on the closed simplex and equals k only at the center. There are at most N^s bins, so they contribute $O(z^{2k}e^{(k-1-c'_\eta)z}) = o(e^{(k-1)z})$. This proves (2).

Step 5: part (3). Equate the 2 expressions in (2) and let $N \rightarrow \infty$. This gives $k = D_k(2\pi)^{s/2}(\det \Sigma_k)^{1/2}$, that is, $D_k = k \det(2\pi\Sigma_k)^{-1/2}$. As a check, $\det((k^2/2)(I + J)) = (k^2/2)^s k$, which gives $D_k = k^{k+1/2}(4\pi)^{-s/2}$ again. $\square$

Proof of Theorem 3.8. At $p = 1/d$ the chain is a sequence of independent uniform states, and the claim follows from the multinomial formula: the move multiplies the probability by $n_b/(n_a + 1) > 1$. For $1/d < p < 1$ we have $\sigma > 0$ and $v = r/\sigma > 0$.

Step 1: an integral form. Writing $M! = \int_0^\infty e^{-t} t^M dt$ in the uniform case of Proposition 2.2(a) gives

$$P_N(n) = \frac{\sigma^{N-1}}{dv} \int_0^\infty e^{-t} \prod_i B_{n_i}(vt) dt, \quad B_n(y) = \sum_{m=1}^n \binom{n-1}{m-1} \frac{y^m}{m!}, \quad (17)$$

where the product runs over the positive coordinates.

Step 2: log-concavity. We show that for each $y > 0$ the sequence $B_n(y)$, $n \geq 1$, is strictly log-concave. Write

$$\frac{B_n(y)}{y} = b_{n-1}, \quad b_j = \sum_{i=0}^j \binom{j}{i} a_i, \quad a_i = \frac{y^i}{(i+1)!}, \quad c_j = \sum_{i=0}^j \binom{j}{i} a_{i+1}.$$

Pascal's identity gives $b_{j+1} = b_j + c_j$. The ratio c_j/b_j is the mean of the strictly decreasing sequence $f_i = a_{i+1}/a_i = y/(i+2)$ under the weights $\pi_i = \binom{j}{i} a_i$, $0 \leq i \leq j$, normalized. Let $\pi'_i = \binom{j+1}{i} a_i$, $0 \leq i \leq j+1$, be the weights for $j+1$, and put $\pi_{j+1} = 0$. On $0 \leq i \leq j$ the ratio $\pi'_i/\pi_i = (j+1)/(j+1-i)$ increases in i . So for $i < l$ we have $\pi'_i \pi_l - \pi'_l \pi_i \leq 0$, with strict inequality when $l = j+1$. Hence

$$\sum_i \pi'_i f_i \sum_l \pi_l - \sum_i \pi'_i \sum_l \pi_l f_l = \sum_{i < l} (\pi'_i \pi_l - \pi'_l \pi_i) (f_i - f_l) < 0,$$

since $f_i > f_l$ for $i < l$. So the mean decreases strictly, $c_{j+1}/b_{j+1} < c_j/b_j$, and $b_{j+1}/b_j = 1 + c_j/b_j$ decreases strictly with j . This is the strict log-concavity.

Step 3: the transfer. Let $b \geq a + 2$. It follows that $B_{a+1}(y)/B_a(y) > B_b(y)/B_{b-1}(y)$, that is, $B_{a+1}(y)B_{b-1}(y) > B_a(y)B_b(y)$. Apply this for $y = vt > 0$ in (17) and integrate. So the move strictly increases $P_N(n)$. Repeated moves reach a balanced bin from any bin with the same support, and all balanced bins with this support have the same probability by symmetry. This proves the theorem. $\square$

Proof of Corollary 3.9. For large N , $T \leq CL$ forces $r \rightarrow 0$ and so $p > 1/d$. By Theorem 3.8 every global mode is a vertex or a balanced bin, and all balanced bins with the same number of positive coordinates have the same probability. So we compare the ratios $C_{N,k}/V_N = S_n(u)$, $2 \leq k \leq d$, with 1 and with each other. Fix $0 < c < 1$.

Case $T \leq cL$. The ratio $S_n(u)$ increases with u , and u increases with T . So it is at most its value at $T = cL$, which is $\exp(-(k-1)(1-c)L + O(\log L)) \rightarrow 0$ by Theorem 3.1(i). So the vertices are the only modes, and $p > p_{N,d}$.

Case $cL \leq T \leq CL$. Theorem 3.1(i) gives $\log(C_{N,k}/V_N) = (k-1)(\Psi_N(T) - a_k) + o(1)$, uniformly. Fix $\delta > 0$, with $\delta < (a_{d-1} - a_d)/2$ when $d \geq 3$. This is possible since $a_2 > \dots > a_d$. If $\Psi_N(T) \leq a_d - \delta$, every face ratio is at most $e^{-(k-1)\delta + o(1)} < 1$, so only the vertices are modes. If $\Psi_N(T) \geq a_d + \delta$, the full center beats the vertices, and it beats a balanced bin with $2 \leq k < d$ positive coordinates by

$$(d-k)(\Psi_N(T) - a_d) + (k-1)(a_k - a_d) + o(1) > 0.$$

If $|\Psi_N(T) - a_d| < \delta$, a proper face with $k \geq 2$ states has $(k-1)(\Psi_N(T) - a_k) + o(1) \leq -(k-1)\delta + o(1) < 0$, since $a_k - a_d > 2\delta$. So it loses to the vertices, and the comparison of the vertices with the full centers is decided by the unique crossing $p_{N,d}$ of Theorem 3.2. In each subcase the conclusion agrees with (i) to (iii).

The last claim follows from (5) and $p_{N,k} = 1 - (d-1)z_{N,k}/N$, since $a_2 > \dots > a_d$. $\square$

B Proofs for Sections 4 and 5

B.1 The start weights

We prove Lemma 4.1. Throughout $1/d < p < 1$, so $\sigma = p - r > 0$ and $v = r/\sigma = u/(1-u)$. The refresh form in Proposition 2.2(a) is linear in μ , and with $\mu = \delta_a$ it gives $p^{N-1}S_n^{(a)}(u)$. So, with

sums over $1 \leq m_b \leq n_b$ for $b \in F$ and $M = \sum_b m_b$,

$$\omega_a(n) = \mathbb{E}_w \left[\frac{m_a}{M} \right], \quad w(m) \propto \frac{M!}{\prod_b m_b!} v^M \prod_{b \in F} \binom{n_b - 1}{m_b - 1}. \quad (18)$$

In words, $\omega_a(n)$ is the mean fraction of the blocks of the refresh form that carry state a .

Proof of Lemma 4.1. (1) Exchanging the letters a and b is a bijection from the words with counts n that start with a to those that start with b . It keeps the counts, since $n_a = n_b$, and the number of changes, so $S_n^{(a)} = S_n^{(b)}$. If k divides N , a balanced n has equal coordinates, and the k weights are equal and sum to 1.

(2) We may take $N \geq N_0$ so large that $p > 1/d$ and $\sigma \geq \frac{1}{2}$. Let n be balanced and $\nu = \lfloor N/k \rfloor$, so each n_b is ν or $\nu + 1$, and $\nu \geq N/(2k)$.

Step 1: reduction to 2 values. By (1) the weights take a value ω_+ where $n_b = \nu + 1$ and a value ω_- where $n_b = \nu$. We show that $0 \leq \omega_+ - \omega_- \leq \delta$ with $\delta = \frac{2}{\nu}(\mathbb{E}_w M + \mathbb{E}_w M^2)$. Since the weights average to $1/k$, each is then within δ of $1/k$, and $|\sum_a \mu_a \omega_a(n) - \mu(F)/k| \leq \mu(F)\delta$.

Step 2: pairing. Let $n_a = \nu + 1$ and $n_b = \nu$, and let m' be m with m_a and m_b exchanged, with $w(m') = 0$ if m' is not admissible. All factors of $w(m)$ other than $\binom{\nu}{m_a-1} \binom{\nu-1}{m_b-1}$ are symmetric in m_a and m_b . From $\binom{\nu}{j} / \binom{\nu-1}{j} = \nu/(\nu-j)$ we get $w(m') = w(m)(\nu - m_a + 1)/(\nu - m_b + 1)$ when $m_a \leq \nu$, and $w(m') = 0$ when $m_a = \nu + 1$. In both cases $w(m) - w(m') = w(m)(m_a - m_b)/(\nu - m_b + 1)$ when $m_a > m_b$. The map $m \mapsto m'$ matches the admissible m with $m_a < m_b$ to the admissible m with $m_b < m_a \leq \nu$. Pairing m with m' in (18) gives, with $Z = \sum_m w(m)$,

$$\omega_a - \omega_b = \frac{1}{Z} \sum_{m_a > m_b} (w(m) - w(m')) \frac{m_a - m_b}{M} = \frac{1}{Z} \sum_{m_a > m_b} w(m) \frac{(m_a - m_b)^2}{M(\nu - m_b + 1)} \geq 0.$$

Here $(m_a - m_b)^2/M \leq M$. If $m_b \leq \nu/2$ the last factor is at most $2/\nu$. If $m_b > \nu/2$, then $M > \nu/2$ and the term is at most $w(m)M < w(m)2M^2/\nu$. This proves the bound δ .

Step 3: the moments. It remains to bound $\mathbb{E}_w M$ and $\mathbb{E}_w M^2$. Since $(m-1)!(\binom{n_b-1}{m-1})/n_b^{m-1} = \prod_{j < m} (1 - j/n_b)_+$, we have $w(m) \propto c(m)A(m)$ for all m with every $m_b \geq 1$, where

$$c(m) = M! \prod_b \frac{(n_b v)^{m_b}}{m_b! (m_b - 1)!}, \quad A(m) = \prod_b \prod_{j < m_b} \left(1 - \frac{j}{n_b}\right)_+ \in [0, 1],$$

and Lemma A.2 gives $1 - A(m) \leq \sum_b m_b^2/(2n_b) \leq M^2/\nu$. Write $M! = \int_0^\infty e^{-t} t^M dt$. Since $\sum_{m \geq 1} y^m/(m!(m-1)!) = y\Phi(y)$, this gives $\mathbb{E}_c[(M+1)(M+2)] = \langle t^2 \rangle$, where $\langle \cdot \rangle$ is the mean under the density proportional to $e^{-t} t^k \prod_b \Phi(n_b v t)$, as in the proof of Theorem 6.2. The integration by parts there, with t^{k+2} in place of t^{k+1} , gives $\langle t^2 \rangle = (k+2)\langle t \rangle + \sum_b \langle t \psi(n_b v t) \rangle$. Put $c_1 = \sqrt{v} \sum_b \sqrt{n_b}$. That proof gives $\langle t \rangle \leq 2(k+1) + c_1^2$, and $\psi(x) \leq \sqrt{x}$ and $\langle t^{3/2} \rangle \leq \langle t^2 \rangle^{3/4}$ give $X = \langle t^2 \rangle \leq (k+2)(2k+2+c_1^2) + c_1 X^{3/4}$. If $c_1 X^{3/4} \geq X/2$, then $X \leq 16c_1^4$, and otherwise $X \leq 2(k+2)(2k+2+c_1^2)$. So $X \leq C_2(1 + c_1^2)^2$, where C_2 depends only on k . Here $c_1^2 \leq kNv = kT/\sigma \leq 2kT \leq 2kCL$, so $\mathbb{E}_c M^2 \leq \mathbb{E}_c[(M+1)(M+2)] \leq \nu/2$ for $N \geq N_0$. Since $0 \leq A \leq 1$ and $\mathbb{E}_c[A] \geq 1 - \mathbb{E}_c M^2/\nu \geq \frac{1}{2}$, and $M \leq M^2$,

$$\mathbb{E}_w M \leq \mathbb{E}_w M^2 = \frac{\mathbb{E}_c[M^2 A]}{\mathbb{E}_c[A]} \leq \frac{\mathbb{E}_c M^2}{1 - \mathbb{E}_c M^2/\nu} \leq 2C_2(1 + 2kT)^2.$$

With $\nu \geq N/(2k)$ this gives $\delta \leq C_1(1 + T)^2/N$. $\square$

B.2 Confinement and the field

The next lemma is the Gaussian profile of the ratio to a vertex near the center of a face. We use it for Theorem 5.2.

Lemma B.1 (Gaussian confinement). *Fix a support S with $|S| = k \geq 2$ and a compact set $\mathcal{J}$ of values of t , and put $T = L - \frac{1}{2} \log L + t$. For a positive count vector n on S put $y = \sqrt{L}(n/N - \frac{1}{k}\mathbf{1})$, a vector in the hyperplane $\{y \in \mathbb{R}^S : \sum_i y_i = 0\}$.*

(a) *Uniformly for $t \in \mathcal{J}$ and y in a compact subset of this hyperplane,*

$$\log S_n(u) = (k-1)(t - a_k) - \frac{k^2}{4}|y|^2 + o(1).$$

(b) *For every $R > 0$, uniformly for $t \in \mathcal{J}$ and all n with $|y| > R$,*

$$\log S_n(u) \leq (k-1)(t - a_k) - \frac{k^2}{4}R^2 + o(1).$$

Proof. (a) Put $\alpha = n/N = \frac{1}{k}\mathbf{1} + y/\sqrt{L}$. For $|y|$ bounded, $\alpha_i \geq 1/(2k)$ for large N , and Taylor expansion of $\sqrt{1/k+x} = k^{-1/2} + \frac{\sqrt{k}}{2}x - \frac{k^{3/2}}{8}x^2 + O(x^3)$, with $\sum_i y_i = 0$, gives

$$\left(\sum_i \sqrt{\alpha_i}\right)^2 = k - \frac{k^2}{4L}|y|^2 + O(L^{-3/2}), \quad D(\alpha) = D_k(1 + o(1)).$$

Insert these into Theorem 3.3, with $\varepsilon = 1/(2k)$, $c = \frac{1}{2}$, $C = 2$ and $z = Nu = T/p$, where $z - T = O(L^2/N)$. Then $(A^2 - 1)z = (k-1)z - \frac{k^2}{4}|y|^2 + o(1)$ and $z + \frac{1}{2} \log z - L = t + \frac{1}{2} \log(z/L) + o(1) = t + o(1)$. Since $\log D_k = -(k-1)a_k$, this gives (a).

(b) Starting from n , move one unit from a coordinate b to a coordinate a with $n_b \geq n_a + 2$, and repeat. The moves keep the support and end at a balanced vector, and by Theorem 3.8 each move strictly increases $S_n(u)$. A move changes $|y|^2$ by $\frac{2L}{N^2}(n_a - n_b + 1) < 0$ and changes y by a vector of length $\sqrt{2L}/N$. The balanced end has $|y| = O(\sqrt{L}/N)$. So if $|y| > R$ at the start, the first vector on the path with $|y| \leq R$ has $|y| \geq R - \sqrt{2L}/N = R - o(1)$, and by (a), applied on the compact set $\{|y| \leq R\}$, its value of $\log S_n$ is at most $(k-1)(t - a_k) - k^2 R^2/4 + o(1)$, uniformly. The starting vector has a smaller value. $\square$

Proof of Theorem 5.2. Fix compact sets of t and h , and a support S with $|S| = k \geq 2$.

Step 1: the best bin on S . Relative to V_N , a bin with support S has weight $e^{h \cdot \alpha} S_n(u)$ with $\alpha = n/N$. Choose R with $k^2 R^2/4 > \max_i h_i - \bar{h}_S + 1$ on the compact set. Since $h \cdot \alpha \leq \max_{i \in S} h_i$, Lemma B.1(b) shows that bins with $|y| > R$ have log weight at most $\ell_S(t, h) - 1 + o(1)$. On $|y| \leq R$ we have $h \cdot \alpha = \bar{h}_S + O(R/\sqrt{L})$, and Lemma B.1(a) gives the log weight $\ell_S(t, h) - \frac{k^2}{4}|y|^2 + o(1)$. A balanced bin has $|y| = O(\sqrt{L}/N)$. So the largest weight on S is $e^{\ell_S(t, h) + o(1)}$, while a vertex i has weight exactly e^{h_i} .

Step 2: the modes. There are finitely many supports. This proves (10), and a positive gap to a lower score excludes that support for large N . If the maximizers on a selected support had $|y| \geq \eta > 0$ along a subsequence, their log weight would be at most $\ell_S - k^2 \eta^2/4 + o(1)$, below that of a balanced bin. This proves the location statement.

Step 3: the lines. Among supports of size k the largest $\bar{h}_S$ is the average of the k largest fields, which gives (11). If k_1 wins at t_1 and k_2 at $t_2 > t_1$, adding the 2 comparisons gives $(k_1 - k_2)(t_2 - t_1) \leq 0$. So the winning size is nondecreasing away from ties.

Step 4: every intermediate size. To realize a given $2 \leq k < d$, set $h_1 = \dots = h_k = 0$ and $h_{k+1} = \dots = h_d = -H$, and take $t > a_k$ so close to a_k that $t < a_{k-1}$ if $k > 2$. Then the k -face score is positive, the largest vertex score is 0, and every face with $2 \leq j < k$ states scores at most $(j-1)(t-a_j) < 0$. A face with $j > k$ states scores at most $(j-1)(t-a_j) - H(j-k)/j$, and another face with k states at most $(k-1)(t-a_k) - H/k$, so both lose once H is large. The comparisons are strict, so they hold on an open interval of t , and small changes make the fields strictly ordered without removing the phase. $\square$

Proof of Corollary 5.6. Since $\frac{1}{k} \sum_{i=1}^k (i-1)^2 = (k-1)(2k-1)/6$, the lines (11) are $\ell_k(t) = (k-1)t + c_k - H(k-1)(2k-1)/6$, also for $k=1$, and $\ell_{k+1}(t) - \ell_k(t) = t - t_k$. Extend c_k to the smooth function $c(x) = (x + \frac{1}{2}) \log x - \frac{x-1}{2} \log(4\pi)$ for $x \geq 1$. Then $0 < c''(x) = (x - \frac{1}{2})/x^2 \leq \frac{1}{2}$, and integrating twice over unit intervals gives $c_{k+2} - 2c_{k+1} + c_k \leq \frac{1}{2}$. Hence

$$t_{k+1} - t_k = \frac{2H}{3} - (c_{k+2} - 2c_{k+1} + c_k) > 0.$$

For $t_{k-1} < t < t_k$, summing the differences $\ell_{j+1}(t) - \ell_j(t) = t - t_j$ shows that line k strictly exceeds every other line. The fields are strictly decreasing, so the unique best support of size k is $\{1, \dots, k\}$, and Theorem 5.2 gives the conclusion for finite N . The arithmetic in this proof is checked in Lean. $\square$

C The majorant and the crossover at $N/(\log N)^2$

This appendix proves Lemma 6.1, Theorem 6.2 and the results behind Remark 6.3. The start is uniform.

C.1 The majorant

Proof of Lemma 6.1. By definition, $B_a(w)/w = \sum_{j=0}^{a-1} \binom{a-1}{j} w^j / (j+1)!$. So the coefficient of $(aw)^j / [j!(j+1)!]$ in $B_a(w)/w$ is $\binom{a-1}{j} j! / a^j = \prod_{h=1}^j (1 - h/a)_+$, also for $j \geq a$, where it is 0. By Lemma A.2 its deficit from 1 lies between 0 and $\sum_{h \leq j} h/a = j(j+1)/(2a)$. Since $\sum_j j(j+1)x^j / [j!(j+1)!] = x\Phi(x)$ (Lemma A.1(i)), we get

$$0 \leq \Phi(aw) - \frac{B_a(w)}{w} \leq \frac{1}{2a} \sum_{j \geq 0} \frac{j(j+1)(aw)^j}{j!(j+1)!} = \frac{w}{2} \Phi(aw). \quad \square$$

Proof of Theorem 6.2. Step 1: an integral form. Let $0 < u < 1$ and $v = u/(1-u)$. Put $M! = \int_0^\infty e^{-t} t^M dt$ into (3) and sum over each m_i separately. All terms are nonnegative, so

$$S_n(u) = \frac{(1-u)^{N-1}}{v} \int_0^\infty e^{-t} \prod_{i=1}^k B_{n_i}(vt) dt, \quad (19)$$

with B_a as in Lemma 6.1.

Step 2: comparison of the factors. For $t > 0$ put $f_i(t) = B_{n_i}(vt)/(vt \Phi(n_i vt)) \in [0, 1]$. By Lemma 6.1 we have $f_i(t) = (1 - t_i)_+$ with $0 \leq t_i \leq vt/2$, so Lemma A.2 gives $0 \leq 1 - \prod_i f_i(t) \leq kv/2$. Write $\langle g \rangle$ for the mean of $g(t)$ under the probability density on $(0, \infty)$ proportional to $e^{-t} t^k \prod_i \Phi(n_i vt)$. Then (19) reads $S_n(u) = \mathcal{Q}_n(u) \langle \prod_i f_i \rangle$, and so

$$0 \leq 1 - \frac{S_n(u)}{\mathcal{Q}_n(u)} = \left\langle 1 - \prod_i f_i \right\rangle \leq \frac{kv}{2} \langle t \rangle.$$

The bound $\Phi(x) \leq e^{2\sqrt{x}}$ of Section 2 bounds the density by a polynomial times $\exp(-t + 2\sqrt{vt} \sum_i \sqrt{n_i})$, so all moments below exist.

Step 3: the mean of t . We use $\psi(x) = x\Phi'(x)/\Phi(x)$ from Lemma A.1. Since $\frac{d}{dt} \log \Phi(n_i vt) = \psi(n_i vt)/t$,

$$\frac{d}{dt} \left[e^{-t} t^{k+1} \prod_i \Phi(n_i vt) \right] = e^{-t} t^k \prod_i \Phi(n_i vt) \left(k + 1 - t + \sum_i \psi(n_i vt) \right).$$

The function in square brackets vanishes at $t = 0$ and tends to 0 as $t \rightarrow \infty$, so integration gives $\langle t \rangle = k + 1 + \sum_i \langle \psi(n_i vt) \rangle$. By Lemma A.1, $\psi(x) \leq \sqrt{x}$. With $\langle \sqrt{t} \rangle \leq \sqrt{\langle t \rangle}$ and $c = \sqrt{v} \sum_i \sqrt{n_i}$ this gives $\langle t \rangle \leq k + 1 + c\sqrt{\langle t \rangle}$. Since $c\sqrt{\langle t \rangle} \leq (c^2 + \langle t \rangle)/2$, we get $\langle t \rangle \leq 2(k + 1) + v(\sum_i \sqrt{n_i})^2$. This proves the first bound in (13), and Cauchy–Schwarz, $(\sum_i \sqrt{n_i})^2 \leq kN$, gives the second. $\square$

C.2 The crossover

Fix integers $k_1 \geq 2$ and $\ell \geq 0$, and put $k = k_1 + \ell$. We take k_1 large counts $n_1, \dots, n_{k_1}$ and ℓ rare counts $a_1, \dots, a_\ell$, all positive, and write $S_{(n,a)}$ for the ratio S of this count vector. For $\ell = 0$ there are no rare counts, and the sums and products over j below are empty. Put

$$N = \sum_i n_i + \sum_j a_j, \quad M = \sum_i n_i, \quad L = \log N, \quad \beta_i = \frac{n_i}{M}, \quad \rho_j = \frac{a_j L^2}{N}.$$

The β_i and ρ_j always refer to these actual counts. In this appendix

$$A = \sum_{i=1}^{k_1} \sqrt{\beta_i}, \quad g = A^2 - 1, \quad \eta = \frac{k_1 - 1}{2} + 2\ell, \quad c = \frac{k - 1}{g},$$

$$D_{k_1}(\beta) = \frac{A^{k_1/2+1} (\prod_i \beta_i)^{-3/4}}{(4\pi)^{(k_1-1)/2}}, \quad \gamma_0 = \frac{(2\eta + 1)(2\eta + 3)}{16A^2} - \frac{3}{16A} \sum_{i=1}^{k_1} \beta_i^{-1/2}.$$

The constant $D_{k_1}(\beta)$ is $D(\beta)$ of Theorem 3.3 on k_1 states, so it has the Kirchhoff form of Proposition 3.6, $D_{k_1}(\beta) = A^2 (4\pi)^{-(k_1-1)/2} \sqrt{\text{tree}(\beta)} / \prod_i \beta_i$. We use $\psi(x) = x\Phi'(x)/\Phi(x)$ and its properties from Lemma A.1.

Theorem C.1 (a crossover between neighboring simplex strata). *Fix $\varepsilon > 0$ and $R < \infty$. Assume $\beta_i \geq \varepsilon$ and $0 < \rho_j \leq R$. For $z > 0$ put $x_j = A^2 a_j z^2 / N$ and*

$$\mathcal{E}(z) = -gz \sum_{j=1}^{\ell} \frac{a_j}{N} + \frac{1}{A^2 z} \left(\eta \sum_{j=1}^{\ell} \psi(x_j) + 2 \sum_{1 \leq i < j \leq \ell} \psi(x_i) \psi(x_j) \right) + \frac{\gamma_0}{z},$$

so that $\mathcal{E}(z) = \gamma_0/z$ when $\ell = 0$. Let $0 < c_0 < C_0 < \infty$ be fixed. Then, uniformly for such count vectors and $c_0 L \leq z \leq C_0 L$,

$$S_{(n,a)}(z/N) = D_{k_1}(\beta) A^{2\ell} \frac{z^\eta e^{gz}}{N^{k-1}} \prod_{j=1}^{\ell} \Phi(x_j) e^{\mathcal{E}(z)} \left(1 + O\left(L^{-2} + \frac{L^2}{N}\right) \right). \quad (20)$$

Also $|\mathcal{E}(z)| \leq C_1(1+R)/L$ there, where C_1 depends only on $\varepsilon, c_0, C_0, k_1$ and ℓ . So (20) also holds without the factor $e^{\mathcal{E}(z)}$ if the error is replaced by $O(L^{-1} + L^2/N)$.

For large N , the right side of (20) without its error factor equals 1 at exactly one ζ_N in $[(k-1)L/(2(k_1-1)), (k-1)L/\varepsilon]$. The unique positive solution $u_{n,a}$ of $S_{(n,a)}(u) = 1$ satisfies

$$Nu_{n,a} = \zeta_N + O\left(L^{-2} + \frac{L^2}{N}\right) \quad (21)$$

and

$$\begin{aligned} Nu_{n,a} = cL - \frac{\eta}{g} \log L - \frac{1}{g} \left\{ \log(D_{k_1}(\beta)A^{2\ell}) + \eta \log c + \sum_{j=1}^{\ell} \log \Phi(A^2 c^2 \rho_j) \right\} \\ + O\left(\frac{\log L}{L} + \frac{L^2}{N}\right). \end{aligned} \quad (22)$$

The constants in the errors of (20) depend only on ε , R , k_1 , ℓ , c_0 and C_0 . Those in (21) and (22), and the meaning of “large N ”, depend only on ε , R , k_1 and ℓ .

In words, each rare count multiplies the profile of the large counts by $A^2 z^2/N$ (through $A^{2\ell}$, z^η and N^{k-1}) and by a Bessel factor $\Phi(x_j)$, and the 3 terms of $\mathcal{E}$ give all corrections of relative order $1/L$. At $u = z/N$ we have $T = Nr = z/(1 + (d-1)z/N) = z + O(L^2/N)$. The logarithms of the right sides have bounded derivatives in z , and all errors contain L^2/N . So the theorem holds verbatim with T in place of z and of $Nu_{n,a}$.

Proof. Let $u = z/N$ with $c_0L \leq z \leq C_0L$, and take N large. Put $v = u/(1-u)$, $Y = Mv$, $w_0 = A\sqrt{Y}$, $q_0 = \eta + 3/2 = 2\ell + 1 + k_1/2$, $s_a = \sum_j a_j/N$ and $\xi_j = a_j v w_0^2 = A^2 a_j M v^2$. Then $s_a \leq \ell R/L^2$ and $Y = z(1-s_a)/(1-z/N)$, so $Y \asymp L$. Also $\xi_j \leq X = 4k_1 C_0^2 R$. When $\ell = 0$ we have $M = N$ and $s_a = 0$, and the steps that involve the ξ_j and X are empty. Note that $A^2 = 1 + 2 \sum_{i < i'} \sqrt{\beta_i \beta_{i'}}$ lies in $[1 + 2\varepsilon, k_1]$.

Step 1: the Bessel integral. By Theorem 6.2, with relative error $O(v + Nv^2) = O(L^2/N)$, the ratio $S_{(n,a)}(u)$ equals $\mathcal{Q}_{(n,a)}(u)$. Writing $\Phi(\beta_i Y t) = I_1(2\sqrt{\beta_i Y t})/\sqrt{\beta_i Y t}$ for the large counts, this is

$$\frac{(1-u)^{N-1} v^{\ell-1}}{\prod_{i=1}^{k_1} n_i} \int_0^\infty e^{-t} t^\ell \prod_{i=1}^{k_1} \left[\sqrt{\beta_i Y t} I_1(2\sqrt{\beta_i Y t}) \right] \prod_{j=1}^{\ell} \Phi(a_j v t) dt. \quad (23)$$

Substitute $t = w^2$. The integrand becomes

$$2w^{2\ell+1} e^{-w^2} \prod_{i=1}^{k_1} \left[\sqrt{\beta_i Y} w I_1(2\sqrt{\beta_i Y} w) \right] \prod_{j=1}^{\ell} \Phi(a_j v w^2).$$

Put $F(w) = w^{q_0} \prod_j \Phi(a_j v w^2)$, so that $F(w_0) = A^{q_0} Y^{q_0/2} \prod_j \Phi(\xi_j)$.

Step 2: the main part. On $I = \{|w - w_0| \leq w_0/2\}$ the Bessel arguments $2\sqrt{\beta_i Y} w$ are at least $A\sqrt{\varepsilon} Y$. The expansion of I_1 in Section 2, and $-w^2 + 2A\sqrt{Y} w = A^2 Y - (w - w_0)^2$, then show that on I the integrand equals

$$\frac{2(\prod_i \beta_i)^{1/4} Y^{k_1/4}}{(4\pi)^{k_1/2}} e^{A^2 Y - (w - w_0)^2} F(w) \left(1 - \frac{b_Y}{w} + O(Y^{-2}) \right), \quad b_Y = \frac{3}{16\sqrt{Y}} \sum_i \beta_i^{-1/2}.$$

We apply Lemma A.3 with $c_j = a_j v$, once with $q = q_0$ to F and once with $q = q_0 - 1$ to $F(w)/w$ for the term b_Y/w . For the term $O(Y^{-2})$ we use its bound $F(w_0 + y) = O(F(w_0))$. Since

$b_Y/w_0 = O(1/Y)$, the integral over I equals

$$\frac{2\sqrt{\pi}(\prod_i \beta_i)^{1/4} Y^{k_1/4}}{(4\pi)^{k_1/2}} e^{A^2 Y} F(w_0) \left(1 + \frac{\mu_{q_0}}{4A^2 Y} - \frac{3}{16AY} \sum_i \beta_i^{-1/2} + O(Y^{-2}) \right),$$

with μ_{q_0} as in the lemma.

Step 3: the tails. The part outside I is negligible. The bounds $I_1(x) \leq e^x$ and $\Phi(x) \leq e^{2\sqrt{x}}$ bound the integrand there by a polynomial in w and Y times $\exp(A^2 Y - (w - w_0)^2 + 2\delta w)$, where $\delta = \sum_j \sqrt{a_j v}$, and $\delta w_0 = \sum_j \sqrt{\xi_j}$ is bounded. Completing the square gives $-(w - w_0 - \delta)^2 + 2\delta w_0 + \delta^2$, and outside I we have $|w - w_0 - \delta| \geq 3w_0/8$ for large N . Hence this part is at most a power of Y times $e^{A^2 Y - 9A^2 Y/64}$, which is $O(L^{-2})$ relative to the main term.

Step 4: collecting the constants. Since $v^{\ell-1}/\prod_i n_i = Y^{\ell-1} M^{-(k-1)}/\prod_i \beta_i$ and $\ell - 1 + k_1/4 + q_0/2 = \eta$, the constants collect to $D_{k_1}(\beta) A^{2\ell} Y^\eta M^{-(k-1)}$. Since $4q_0 - 6 = 4\eta$ and $q_0(q_0 - 1) = (2\eta + 1)(2\eta + 3)/4$, the formula for μ_q in Lemma A.3 gives

$$\frac{\mu_{q_0}}{4A^2} - \frac{3}{16A} \sum_i \beta_i^{-1/2} = \frac{1}{A^2} \sum_j \xi_j + \Theta(\xi), \quad \Theta(\xi) = \frac{1}{A^2} \left(\eta \sum_j \psi(\xi_j) + 2 \sum_{i < j} \psi(\xi_i) \psi(\xi_j) \right) + \gamma_0.$$

Note that $\sum_j \xi_j/(A^2 Y) = v \sum_j a_j$. Both $\sum_j \xi_j/A^2$ and $\Theta(\xi)$ are bounded, so the bracket in Step 2 equals $\exp(v \sum_j a_j + \Theta(\xi)/Y)(1 + O(Y^{-2}))$. We have proved

$$S_{(n,a)}(u) = D_{k_1}(\beta) A^{2\ell} \frac{Y^\eta}{M^{k-1}} (1-u)^{N-1} e^{v(A^2 M + \sum_j a_j)} \prod_j \Phi(\xi_j) e^{\Theta(\xi)/Y} \left(1 + O\left(L^{-2} + \frac{L^2}{N}\right) \right). \quad (24)$$

Step 5: from (M, Y) to (N, z) . Now use $u = z/N$. Since $A^2 M + \sum_j a_j = N(A^2 - g s_a)$ and $vN = z + O(L^2/N)$, the exponent is

$$v\left(A^2 M + \sum_j a_j\right) + (N-1) \log(1-u) = gz - gzs_a + O(L^2/N).$$

Also $Y^\eta M^{-(k-1)} = z^\eta N^{-(k-1)}(1 + O(L^{-2} + L/N))$ and $\xi_j = x_j(1 - s_a)(1 - z/N)^{-2} = x_j(1 + O(L^{-2} + L/N))$. Since $(\log \Phi)'(x) = \psi(x)/x \leq 1/2$, replacing ξ_j by x_j in Φ changes the product by a relative $O(L^{-2} + L/N)$. Since $0 \leq \psi' \leq 1$, replacing $\Theta(\xi)/Y$ by $\Theta(x)/z$ changes it by $O(L^{-3} + 1/N)$. Finally $\mathcal{E}(z) = -gzs_a + \Theta(x)/z$, which proves (20).

Step 6: the bound on $\mathcal{E}$. Note that $\psi(x_j) \leq \sqrt{x_j}$, $2\psi(x_i)\psi(x_j) \leq \psi(x_i)^2 + \psi(x_j)^2 \leq x_i + x_j$, and $A \geq 1$. So $|\mathcal{E}(z)| \leq gzs_a + (\eta \sum_j \sqrt{x_j} + (\ell - 1) \sum_j x_j + |\gamma_0|)/z$. Here $gzs_a \leq (k_1 - 1)C_0 \ell R/L$, $x_j = A^2 \rho_j (z/L)^2 \leq k_1 C_0^2 R$, $z \geq c_0 L$ and $\sqrt{R} \leq 1 + R$, and $|\gamma_0|$ is bounded in terms of ε , k_1 and ℓ .

Step 7: the crossing. Apply (20) with $c_0 = (k-1)/(2(k_1-1))$ and $C_0 = (k-1)/\varepsilon$. Since $g \in [2\varepsilon, k_1-1]$, we have $gc_0 - (k-1) \leq -(k-1)/2$ and $gC_0 - (k-1) \geq k-1$. Let $f(z)$ be the logarithm of the right side of (20) without its error factor. Then $f'(z) = g + (\eta + 2 \sum_j \psi(x_j))/z + \mathcal{E}'(z)$, since $x'_j = 2x_j/z$. Also $\mathcal{E}'(z) = O(L^{-2})$, because $gs_a = O(L^{-2})$, $0 \leq \psi' \leq 1$ and the x_j are bounded. So $f' \geq g/2 \geq \varepsilon$ on $[c_0 L, C_0 L]$ for large N . Hence $f(c_0 L) = (gc_0 - (k-1))L + O(\log L)$ is negative and $f(C_0 L) = (gC_0 - (k-1))L + O(\log L)$ is positive, and f has exactly one zero ζ_N in the interval. By (20) the same signs hold for $\log S_{(n,a)}(z/N)$. The polynomial $S_{(n,a)}$ has nonnegative coefficients, zero constant term and the positive coefficient $k!$ at u^{k-1} , so the crossing is unique and lies in the interval. There $|f(Nu_{n,a})| = O(L^{-2} + L^2/N)$, and $f' \geq \varepsilon$ gives (21).

Step 8: the explicit expansion. For (22), take $f(\zeta_N) = 0$ and use $\mathcal{E} = O(L^{-1})$. This gives

$$g\zeta_N + \eta \log \zeta_N = (k-1)L - \log(D_{k_1}(\beta)A^{2\ell}) - \sum_j \log \Phi(A^2 \rho_j (\zeta_N/L)^2) + O(L^{-1}).$$

First $\zeta_N = cL + O(\log L)$. The derivative of each $\log \Phi(A^2 \rho_j (z/L)^2)$ in z is $2\psi/z = O(L^{-1})$, so the sum of logarithms differs from its value at $\zeta_N = cL$ by $O(\log L/L)$. Substituting this into $\log \zeta_N$ and using (21) proves (22). $\square$

Remark C.2. The 3 terms of $\mathcal{E}$ have different sources. Replacing M by N in the exponent $A^2 Y$ gives a factor $\exp(-A^2 z \sum_j a_j/N)$. The Gaussian average of the factors Φ gives back $\exp(z \sum_j a_j/N)$, which is the term $v \sum_j a_j$ in (24). Together they give the first term of $\mathcal{E}$. The second term comes from the same Gaussian average. The factors Φ and the power of w shift the saddle point, and ψ measures this shift. The last term does not depend on the sizes a_j . For $\ell = 0$ it is all of $\mathcal{E}$, and for equal β_i its coefficient γ_0 is then $-(k-1)/8$, the coefficient of $1/z$ in (6). For $\ell = 1$ the first term of $\mathcal{E}$ dominates for large x_1 . For $\ell \geq 2$ the cross terms are also linear in the x_j , since $\psi(x)/\sqrt{x} = I_2(2\sqrt{x})/I_1(2\sqrt{x})$ tends to 1 by [18, Eq. 10.40.1].

Take 2 equal large counts and one rare count, so that $g = 1$ and $\eta = 5/2$, and let $z = \log N$. Then $x_1 = 2\rho$, and the first 2 terms of $z\mathcal{E}$ are $-\rho + 1.25\psi(2\rho)$. They nearly cancel at $\rho = 1$, so the first term alone is not the correction. The ratio of $S_{(n,a)}$ to the right side of (20) without $e^{\mathcal{E}}$ is 1.04 for $\rho = 1$ and $N = 2000$, and with the first term alone it would be 1.19. For $\rho = 8$ it is 0.36, 0.53, 0.63 and 0.70 at $N = 2 \cdot 10^3, 2 \cdot 10^4, 2 \cdot 10^5$ and $2 \cdot 10^6$. With the factor $e^{\mathcal{E}}$ the value for $\rho = 1$ becomes 0.90, and the 4 values for $\rho = 8$ become 0.52, 0.70, 0.79 and 0.85. At $N = 2000$ we have $L = 7.6$, and the terms of order L^{-2} and L^2/N are comparable to γ_0/L , so the value for $\rho = 1$ does not improve. The form (24) does not replace ξ_j by x_j or $Y^\eta M^{-(k-1)}$ by $z^\eta N^{-(k-1)}$. It gives 0.87 for $\rho = 1$, and 0.78, 0.92, 0.96 and 0.97 for $\rho = 8$.

The constant in $O(L^{-2})$ still depends on R , and the proof gives only a crude bound for it. At $N = 10^{12}$, with $z = L$, the quantity L^2 times the logarithm of the ratio for (20) is about $-1.05, -2.6, -14, -36$ and -266 for $\rho = 1/4, 1, 4, 8$ and 32 . For (24) it is about $-0.9, -1.4, -3.5, -6.2$ and -21 . Expanding $(1 - s_a)^{\eta-k+1}$ and $\Phi(x_j(1 - s_a))$ to first order in $s_a = \sum_j \rho_j/L^2$ shows that the difference of the 2 constants is $-(\sum_j \rho_j)(\eta - k + 1 + \sum_j \psi(x_j)) + O(L^{-1} + L^4/N)$. Here this is $-\rho(\frac{1}{2} + \psi(2\rho))$, about -248 for $\rho = 32$. So the first constant grows roughly like $\rho^{3/2}$ in these examples, while the second grows roughly like ρ .

The expansion (22) converges slowly. In the same example, for $\rho = 8$ and $N = 10^6$, it gives 6.58, while $Nu_{n,a} = 14.99$. The root ζ_N of (21) is 14.86, and the root of (24) is 14.97. For $\rho = 1$ the 4 values are 15.41, 17.86, 17.85 and 17.86. The loss is caused by expanding around $z = cL$. Here $c = 2$, while $Nu_{n,a}/L$ is only 1.08 for $\rho = 8$ and 1.29 for $\rho = 1$.

These numbers come from `verify_crossover_factor.py`. For $N \leq 2 \cdot 10^6$ it computes $S_{(n,a)}$ in 3 ways: by the proper-run formula and by a quadrature of (19), both in double precision, and by the sum (3) in 60-digit decimal arithmetic. The values at $N = 10^{12}$ and the crossings come from the quadrature and the sum only. These agree to about 10^{-14} in $\log S$, and the crossings, found by bisection with the sum, agree with the quadrature to about 10^{-7} .

Corollary C.3 (invisibility of subcritical rare counts). *Under the hypotheses of Theorem C.1, if every $a_j = o(N/L^2)$, the sum of the terms $\log \Phi$ in (22) is $o(1)$. So the sizes of the rare counts change the expansion only by $o(1)$. Their number ℓ still changes its leading and logarithmic coefficients.*

Proof. Since $\Phi(0) = 1$ and $0 \leq (\log \Phi)' \leq 1/2$ (Lemma A.1), we have $0 \leq \log \Phi(x) \leq x/2$, and here $x = A^2 c^2 \rho_j \rightarrow 0$. $\square$

The scale $N/(\log N)^2$ needs at least 2 coordinates of order N . It does not apply near a vertex, for example at an endpoint in the 2-state case, where only one coordinate is large and $g = 0$ (Subsection 6.1). The theorem gives no error bound when some ρ_j is unbounded.

D Proofs near a vertex

We use the notation of Subsection 6.1 and prove Theorem 6.4. The first lemma collects what we need about the polynomial H of the rare counts, and the second writes $S_{(b,a)}$ exactly as an average of H .

Lemma D.1 (elasticity and concavity of the rare-count polynomial). *The polynomial H is convex, $\log H$ is concave on $(0, \infty)$, and κ is nondecreasing. Also*

$$\frac{1}{|a|+1} \leq y_a \leq 1, \quad \kappa(y) \leq \ell + \sqrt{\ell|a|y}, \quad \kappa(y) \geq \max\{\ell, \tfrac{1}{2}\sqrt{|a|y}\} \quad (0 < y \leq 1). \quad (25)$$

Proof. Step 1: one factor as a law. Convexity follows directly from the nonnegative coefficients. For one factor B_a and $y > 0$, let J take values in $\{0, \dots, a-1\}$ with $\mathbb{P}(J = j)$ proportional to $\binom{a-1}{j} y^j / (j+1)!$. Since $B_a(y) = \sum_j y \binom{a-1}{j} y^j / (j+1)!$, the elasticity of B_a is $\kappa_a = y B'_a / B_a = \mathbb{E}[J+1]$. The ratio

$$R(j) = \frac{(j+1)\mathbb{P}(J = j+1)}{\mathbb{P}(J = j)} = \frac{y(a-1-j)}{j+2}, \quad R(a-1) = 0,$$

is nonincreasing in j . Summing the ratio identity gives $\mathbb{E}J = \mathbb{E}R(J)$ and $\mathbb{E}J^2 = \mathbb{E}[(J+1)R(J)]$. Therefore

$$\text{Var } J = \mathbb{E}J + \text{Cov}(J, R(J)) \leq \mathbb{E}J.$$

The covariance is nonpositive because, for an independent copy J' , twice the covariance equals $\mathbb{E}[(J - J')(R(J) - R(J'))] \leq 0$.

Step 2: concavity and monotonicity. Since $y^2 B''_a / B_a = \mathbb{E}[J(J+1)]$, we get

$$(\log B_a)'' = \frac{\text{Var } J - \mathbb{E}[J+1]}{y^2} < 0, \quad y \kappa'_a = \text{Var } J \geq 0.$$

Summing over the factors proves the concavity of $\log H$ and the monotonicity of $\kappa = \sum_i \kappa_{a_i}$.

Step 3: the elasticity bounds. The differential equation $y B''_a + y B'_a = a B_a$, which we check on the coefficients, gives

$$ay = \kappa_a^2 + \text{Var } J + (y-1)\kappa_a.$$

Using $0 \leq \text{Var } J \leq \kappa_a - 1$ we get

$$\kappa_a \leq 1 + \sqrt{ay}, \quad \kappa_a^2 + y \kappa_a \geq ay + 1.$$

For $0 < y \leq 1$ the second inequality gives $(\kappa_a + \frac{1}{2})^2 > ay$, and $\kappa_a \geq 1$. So $\kappa_a \geq \max\{1, \sqrt{ay} - 1/2\} \geq \frac{1}{2}\sqrt{ay}$. Summing over the factors proves the elasticity bounds, using Cauchy-Schwarz, $\sum_i \sqrt{a_i} \leq \sqrt{\ell|a|}$, for the upper bound and $\sqrt{\sum_i x_i} \leq \sum_i \sqrt{x_i}$ for the lower one.

Step 4: the root. Finally, $B_a(1) \geq 1$, from its term $m = 1$, and Lemma 6.1 gives

$$B_a(y) \leq y \Phi(ay) \leq y e^{ay/2},$$

where $(j+1)! \geq 2^j$ proves the last inequality term by term. Thus $H(1) \geq 1$, whereas, since $|a|+1 \geq 2$, $H(1/(|a|+1)) \leq (|a|+1)^{-\ell} e^{|a|/(2(|a|+1))} \leq \frac{1}{2} e^{1/2} < 1$. $\square$

Lemma D.2 (an exact compound-binomial representation). *For $0 < u < 1$, put $v = u/(1-u)$. Let G have the gamma distribution with shape 2 and rate 1, let $E_1, \dots, E_{b-1}$ be independent rate-one exponential variables, and let $Z_1, \dots, Z_{b-1}$ be independent Bernoulli variables with parameter u . Take all these variables independent and set*

$$\mathcal{T} = G + \sum_{j=1}^{b-1} Z_j E_j.$$

Then

$$S_{(b,a)}(u) = (1-u)^{|a|} \mathbb{E}[H(v\mathcal{T})], \quad \mathbb{E}e^{s\mathcal{T}} = (1-s)^{-2} \left(1 + \frac{us}{1-s}\right)^{b-1} \quad (s < 1). \quad (26)$$

In particular, set $y = bu^2$, $\kappa = \kappa(y)$ and $s = \kappa/[bu(1-u)]$. Whenever $s < 1$,

$$(1-u)^{|a|} H(y) \leq S_{(b,a)}(u), \quad (27)$$

$$S_{(b,a)}(u) \leq (1-u)^{|a|} H(y) e^{-\kappa} (1-s)^{-2} \left(1 + \frac{us}{1-s}\right)^{b-1}. \quad (28)$$

Proof. By (19) with $N = b + |a|$,

$$S_{(b,a)}(u) = \frac{(1-u)^{b+|a|-1}}{v} \int_0^\infty e^{-t} B_b(vt) H(vt) dt.$$

The integral of $e^{-t} B_b(vt)$ is $\sum_m \binom{b-1}{m-1} v^m = v(1+v)^{b-1}$. After normalization its density is $\sum_{j=0}^{b-1} \binom{b-1}{j} u^j (1-u)^{b-1-j} t^{j+1} e^{-t} / (j+1)!$, a mixture of gamma densities with shape $J+2$, where $J \sim \text{Bin}(b-1, u)$. This is exactly the distribution of $\mathcal{T}$, which proves the identity and its moment generating function. Since $\mathbb{E}\mathcal{T} = 2 + (b-1)u$, we have $v\mathbb{E}\mathcal{T} - bu^2 = u(2-u+bu^2)/(1-u) > 0$, so $v\mathbb{E}\mathcal{T} \geq y$. Jensen's inequality for the convex function H , and the monotonicity of H , prove (27). Concavity of $\log H$ gives, for $w > 0$,

$$H(w) \leq H(y) \exp\left(\frac{\kappa}{y}(w-y)\right).$$

Insert $w = v\mathcal{T}$, note that $\kappa v/y = s$, and use the moment generating function to prove (28). $\square$

Proof of Theorem 6.4. Write $S = S_{(b,a)}$. Put $u_0 = \sqrt{y_a/b}$ and $\kappa_0 = \kappa(y_a)$. Since S increases on $u > 0$, it suffices to show $\log S(u_+) > 0$ and $\log S(u_-) < 0$ at $u_\pm = (1 \pm \delta)u_0$. Note that $\delta \leq 1/2$ forces $\varepsilon \leq 1/(32c_\ell) \leq 1/96$, and that $u_0 \leq \sqrt{1/b} \leq \varepsilon$ since $y_a \leq 1$.

Step 1: the upper point. We have $u_+ \leq \frac{3}{2}\varepsilon < 1/4$. Since $\log H(y_a) = 0$ and $(\log H)' = \kappa/y$, monotonicity of κ gives $\log H(bu_+^2) \geq \kappa_0 \log(bu_+^2/y_a) = 2\kappa_0 \log(1+\delta)$. The lower bound in Lemma D.1 gives $|a|u_0/\kappa_0 \leq 2\sqrt{|a|/b} \leq 2\varepsilon$. Hence, by (27) and $-\log(1-x) \leq x/(1-x)$,

$$\log S(u_+) \geq 2\kappa_0 \log(1+\delta) - \frac{|a|u_+}{1-u_+} \geq \kappa_0 \left(\frac{4}{3}\delta - 4\varepsilon\right) > 0.$$

Here we used $2\log(1+\delta) \geq \frac{4}{3}\delta$ for $\delta \leq \frac{1}{2}$, and $|a|u_+/(1-u_+) \leq \frac{4}{3} \cdot \frac{3}{2}|a|u_0 \leq 4\kappa_0\varepsilon$. The last inequality holds since $\frac{4}{3}\delta = \frac{64}{3}c_\ell\varepsilon > 4\varepsilon$.

Step 2: the lower point. Write $y = bu_-^2$ and $\kappa = \kappa(y)$. The inequalities $y_a \geq 1/(|a| + 1)$, $\delta \leq 1/2$ and the upper bound in (25) imply

$$u_- \leq \varepsilon, \quad bu_- \geq \frac{1}{2\varepsilon}, \quad s = \frac{\kappa}{bu_-(1-u_-)} \leq 2(2\ell + \sqrt{\ell})\varepsilon \leq 4c_\ell\varepsilon \leq \frac{1}{8}.$$

For the bound on s , note that $\kappa/(bu_-) \leq \ell/(bu_-) + \sqrt{\ell|a|y}/(bu_-) \leq 2\ell\varepsilon + \sqrt{\ell|a|/b} \leq (2\ell + \sqrt{\ell})\varepsilon$, and $1 - u_- \geq \frac{1}{2}$. We use $\log(1+x) \leq x$, $-\log(1-s) \leq s/(1-s)$ and $u_-, s \leq 1/4$ in (28), and we drop the nonpositive term $|a|\log(1-u_-)$. This gives

$$\begin{aligned} \log \frac{S(u_-)}{H(y)} &\leq -\kappa + \frac{2s}{1-s} + \frac{bu_-s}{1-s} \\ &= \kappa \left(\frac{1}{(1-u_-)(1-s)} - 1 \right) + \frac{2s}{1-s} \\ &\leq 5\kappa(u_- + s) \leq 25c_\ell\kappa\varepsilon. \end{aligned}$$

Here the equality uses $bu_-s = \kappa/(1-u_-)$. The next inequality uses $(1-u_-)(1-s) \geq \frac{21}{32}$, so the first term is at most $\frac{32}{21}\kappa(u_- + s)$, and $\kappa \geq \ell \geq 1$, so the second is at most $\frac{16}{7}\kappa s$. The last one uses $u_- + s \leq 5c_\ell\varepsilon$. Again by monotonicity of κ ,

$$\log H(y) = - \int_y^{y_a} \frac{\kappa(x)}{x} dx \leq 2\kappa \log(1-\delta) \leq -2\kappa\delta = -32c_\ell\kappa\varepsilon.$$

Thus $\log S(u_-) \leq \kappa(-32c_\ell + 25c_\ell)\varepsilon < 0$. Since S is increasing on $u > 0$, the 2 strict inequalities bracket the crossing. $\square$

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